ASTRA environment-split clustering: cosmological forecasts with emulators and comparison with density split

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Abstract

The distribution of galaxies holds more cosmological information than the standard two-point clustering statistics can capture: the non-linear cosmic web of nodes, filaments, walls and voids is also sensitive to the growth of structure, the matter content, dark energy and neutrinos. We ask how much of this extra information can be recovered with ASTRA, an algorithm that assigns an environment to every point from its neighbours in a Delaunay tessellation of the galaxies and of a random catalogue, without any smoothing scale. We split galaxies and random points by ASTRA environment and measure their auto- and cross-correlations, and compare them with the galaxy two-point function and with density-split clustering, the reference environment statistic. We build the covariances and cosmological derivatives needed for Fisher forecasts from three simulation suites: DESI-like galaxy mocks spanning 85 cosmologies (AbacusSummit), 1786 independent mocks of the official DESI programme, and the Quijote simulations. Along the way, we train a neural-network emulator of all the statistics and use it for an end-to-end inference test. ASTRA environment statistics constrain ω_b, $\omega_{\rm cdm}$, n_s, σ_8, w_0 and w_a about twice as well as the two-point function. They carry as much information as density split, and the two are complementary: combining them improves the constraints further. The original ASTRA cosmic-web classes, tested on Quijote, constrain σ_8 two to three times better than density split. We also describe the main difficulties of this kind of forecast (estimating derivatives from emulator suites, choosing a covariance, handling noisy derivatives) and how we dealt with them.

1 Introduction

Galaxy surveys such as DESI map the large-scale structure of the Universe over volumes large enough that statistical errors are no longer the main limitation. How the galaxies are distributed tells us how structures grew, how much matter there is, and what dark energy and neutrinos do. The question is how much of this information our statistics actually extract.

Most analyses compress the galaxy field into its two-point statistics, the correlation function or the power spectrum. For a Gaussian field this loses nothing, and on large scales it works very well. But gravity makes the late-time Universe strongly non-Gaussian: matter collapses into halos, gathers in filaments and nodes, and empties voids, and peculiar velocities distort the picture in redshift space. The two-point function does not see all of this. Statistics built on the cosmic web recover part of the missing information: [5] showed with Quijote that splitting the density field into nodes, filaments, walls and voids and combining the power spectra of these environments tightens the constraints substantially, most of all on the neutrino mass.

For galaxies, two families of environment statistics have been developed for this purpose. Density-split clustering [1, 2] ranks random points by the smoothed galaxy density around them, splits them into quantiles, and measures the auto-correlation of each quantile and its cross-correlation with the galaxies; combined with a simulation-based emulator [3], it improves ΛCDM constraints by factors of several over the two-point function. Void statistics, combined with galaxy clustering in a forward model of the galaxy–halo connection, also break degeneracies between cosmology and galaxy formation [6].

ASTRA [4] takes a different route. It triangulates the galaxies together with a uniform random catalogue and gives every point an environment from the fraction of galaxies among its Delaunay neighbours: there is no smoothing scale, and the environment adapts to the local density. ASTRA can classify points into voids, sheets, filaments and knots, or rank them continuously.

Goal. Our aim is to quantify how much ASTRA environment statistics improve the constraints on the cosmological parameters (Ω_m through $\omega_{\rm cdm}$, σ_8, the dark-energy parameters w_0 and w_a, and ideally the neutrino mass) over the standard galaxy two-point function, and how they compare with density split when both are measured on the same mocks, with the same random points and the same estimators. To this end we build dedicated covariance and derivative

datasets from suites of mock catalogues, which give Fisher forecasts of the parameter uncertainties, and we train a neural-network emulator of the statistics, which we use for an end-to-end inference test. The neutrino mass turned out to be out of reach with the available simulations (Section 7.2); all the other parameters are covered.

Data. We use three simulation suites, each for what the others cannot provide (Section 2): the AbacusSummit emulator suite of DESI-like galaxy mocks over 85 cosmologies, on which the measurements and the emulator are built; the official DESI mocks, whose 1786 independent boxes give a reliable covariance; and the Quijote simulations, whose paired simulations give clean derivatives. Getting reliable derivatives and covariances was the main difficulty of the work, and we describe how our choices were made and how the results evolved.

Outline. Section 3 presents the two environment estimators, Section 4 the clustering measurements, Section 5 the covariance, the derivatives and the Fisher formalism, Section 6 the successive analysis choices, Section 7 the results (DESI-like mocks, Quijote, ASTRA classes, inference), Section 8 the robustness tests, and Section 9 the discussion. The appendices explain how to reproduce the paper (A), give the mathematical details (B), and retrace the methodological path (C).

2 Simulations and mock catalogues

Table 1 summarises the data used, their versions and their role; exact paths, file names and the lists of catalogues are given in Appendix A.

2.1 AbacusSummit emulator suite

We use the HOD galaxy catalogues built for the DESI emulator programme on top of the AbacusSummit N-body suite [7]. Each simulation evolves 6912^3 particles in a periodic box of $2\,h^{-1}$Gpc; we use the $z = 0.5$ snapshot. Halos are populated with luminous red galaxies (LRG) with the AbacusHOD model [8], whose 12 parameters (central and satellite occupation, velocity bias, environment-based assembly bias) are sampled in the prior of [9]. Every catalogue contains 4×10^6 galaxies, $\bar{n} = 5 \times 10^{-4}\,h^3\mathrm{Mpc}^{-3}$. The suite covers 85 cosmologies.

The catalogues are stored as `c{cosmo}_ph000/seed0/hod{NNN}.fits`. Two properties of this suite drive our methodology:

- **One phase per cosmology.** All cosmologies share the initial conditions `ph000`, with a single HOD seed.
- **One HOD draw per file.** The 500 files `hod000`–`hod499` of a cosmology have *different* HOD parameters (a Latin hypercube for emulator training), and the parameters behind a given index differ from one cosmology to another.

Different HOD files therefore cannot be used as independent realisations of the same model: the scatter between HOD draws reaches $\sim 55\%$ of the amplitude of the density-split monopoles, against $\sim 1\%$ for the cosmological response we want to measure.

Cosmologies and parameters. Besides the fiducial Planck 2018 ΛCDM cosmology (`c000`), we use the AbacusSummit linear derivative grid `c100`–`c113`: seven pairs, each varying one parameter (ω_b, $\omega_{\rm cdm}$, n_s, α_s, w_0, w_0 and w_a together, σ_8) around the fiducial, with h and A_s adjusted to keep the angular size of the sound horizon θ_* and, for `c100`–`c105`, σ_8 fixed. Derivatives with respect to ω_b, $\omega_{\rm cdm}$ and n_s are therefore taken at fixed σ_8. All seven pairs are used to train and validate the emulator. The Fisher forecasts include the four ΛCDM parameters of Table 2, which are those in common with the density-split forecasts of [1] and were the only ones of the first version of this work, and, since the emulator and the derivative grid allow it, the dark-energy parameters w_0 and w_a: six cosmological parameters in total. Going from four to six parameters degrades the marginalised errors on $\omega_{\rm cdm}$ by $\sim 40\%$ and on n_s by $\sim 15\%$, leaves ω_b and σ_8 unchanged, and changes the ASTRA/density-split ratios by less than their uncertainty. The remaining emulator parameters (α_s, $N_{\rm ur}$) are fixed to their fiducial values in the Fisher matrices; all eight are free in the MCMC inference (Section 7.4).

Positions. The line of sight is one axis of the box (z by default, x and y for the subvolume covariance). Redshift-space positions are `X_PERP`, `Y_PERP`, `Z_RSD`; real-space positions use `Z_PERP`. These columns already include the Alcock–Paczynski rescaling of the fiducial cosmology ($q_\perp$, $q_\parallel$, stored in the headers), so the box is $2000/q_\perp$ across and $2000/q_\parallel$ along the line of sight.

2.2 Official DESI AbacusHOD LRG mocks

The emulator suite contains a single phase per cosmology, so its only source of covariance is the 64 subvolumes of one box, measured along three correlated lines of sight (192 samples). This limits the length of

Data set	Version / content	Used	Role
AbacusSummit emulator suite (DESI LRG HOD mocks)	AbacusSummit base boxes $2\,h^{-1}$Gpc, $z = 0.5$, AbacusHOD, `yuan23_prior`; 85 cosmologies, phase `ph000`, seed 0, 500 HOD draws each	$\simeq$ 1040 catalogues: 8 HODs per cosmology (678), 72 fiducial HOD-calibration, 240 near-fiducial (densification), 54 extra; 27 in real space	measurements, emulator training/validation/test, finite differences, regression, single-box covariance (`c000 hod239`)
Official DESI AbacusHOD mocks	`AbacusHOD_mocks/v1`, LRG, $z = 0.5$, model `velbias` (best fit to early DESI LRG data); `CovBox`: 1786 small boxes $500\,h^{-1}$Mpc (`ph3000+`); `CubicBox`: 25 base boxes `c000 ph000--024`	all 1786 small boxes and 25 base boxes	baseline covariance, fiducial point, MCMC test data
Quijote	$1\,h^{-1}$Gpc, 512^3 particles, Rockstar halos, snapshot 3 ($z = 0.5$, `out_3_pid.list`)	4000 `fiducial`; 500 each of `Om`, `Ob2`, `h`, `ns`, `s8` ($\pm$), `Mnu_p/pp/ppp`, `fiducial_ZA`	clean finite-difference Fisher forecasts

Table 1: Data sets used in this work.

Parameter	fiducial	+ / −	step $\theta_+ - \theta_-$
ω_b	0.02237	c100 / c101	0.00089
$\omega_{\rm cdm}$	0.1200	c102 / c103	0.0079
n_s	0.9649	c104 / c105	0.0200
σ_8	0.808	c112 / c113	0.0320

Table 2: AbacusSummit cosmologies of the derivative grid for the four ΛCDM parameters; the pairs `c106`–`c111` vary α_s, w_0 and (w_0, w_a).

the data vectors (Hartlap factor) and, as we found, biases the comparison of the statistics (Section 6). We therefore also use the official DESI AbacusHOD LRG mocks at $z = 0.5$ [8] (`AbacusHOD_mocks/v1`), generated with one best-fit HOD (model `velbias`, fitted to early DESI LRG clustering):

- the 1786 AbacusSummit *small* boxes (`c000`, phases 3000+), of side $500\,h^{-1}$Mpc and independent initial conditions;
- the 25 *base* boxes of `c000` (phases 0–24), of side $2\,h^{-1}$Gpc.

Choices. (i) Each small box is treated exactly like one subvolume of the main analysis: periodic images for the tessellation, density field smoothed on the periodic box, and the *open* Landy–Szalay estimator with geometry randoms for the clustering, so that the covariance corresponds to the same measurement. A periodic small box has no modes larger than its side, whereas a subvolume of a larger box is affected by them (super-sample covariance); the base boxes (64 subvolumes × 25 phases) allow this effect to be measured. (ii) Positions are used as provided, the z coordinate including RSD; only the z line of sight is available. (iii) The best-fit HOD lies inside the prior of the emulator suite and its number density (62 900 galaxies per small box) matches that of our catalogues (62 500 per subvolume), which matters for ASTRA (Section 8). We therefore take it as the fiducial point of the baseline forecasts and of the inference, with this covariance. These mocks contain a single cosmology: they do not provide derivatives.

2.3 Quijote simulations

The derivatives cannot be estimated cleanly by finite differences in the emulator suite, because the HOD differs between cosmologies. The Quijote simulations [12], used by [1] for density-split Fisher forecasts, provide what is missing: for each parameter, 500 pairs of simulations with ± variations of Ω_m (±0.01), Ω_b (±0.002), h (±0.02), n_s (±0.02), σ_8 (±0.015), and simulations with $M_\nu = 0.1, 0.2, 0.4\,$eV, all sharing the initial seeds of the corresponding fiducial realisations, plus thousands of fiducial realisations for the covariance. Each is a periodic box of $1\,h^{-1}$Gpc with 512^3 particles.

Choices. (i) Tracers: all Rockstar host halos at $z = 0.5$ (no subhalos, at least 100 particles). Their number density, $n \simeq 5 \times 10^{-5}\,h^3{\rm Mpc}^{-3}$, is ten times lower than that of the LRG mocks and cannot be raised with these catalogues. Since ASTRA is an adaptive-scale statistic (Section 8), its effective scale, the distance between neighbours $\propto \bar{n}^{-1/3}$, is larger in Quijote ($\simeq 27\,h^{-1}$Mpc) than for the LRGs ($\simeq 13\,h^{-1}$Mpc). Both methods are measured on the same halos, so the comparison of ASTRA and density split remains like-for-like, but at another density: absolute errors and ratios should not be compared directly between the Quijote and AbacusSummit anal-

yses. (ii) Redshift space along z: $z_s = z + v_z/[aH(a)]$, with the Ω_m of each simulation. (iii) The whole box is one volume, with periodic images for the tessellation and the natural estimator of a periodic box (analytic random pairs) for the clustering; density split uses the same smoothing ($R_s = 10\,h^{-1}$Mpc) as for the LRGs. (iv) Derivatives are paired central differences averaged over the realisations shared by the + and − sets; for M_ν, the three-point forward difference $(-3f_0+4f_{0.1}-f_{0.2})/0.2$ with the Zel'dovich-initialised fiducial set (`fiducial_ZA`) for $M_\nu = 0$, as is standard for Quijote. Because the tracers are the same halos in all simulations, these derivatives have no HOD mismatch. The price is realism: halos are not DESI LRGs, so Quijote serves as a controlled comparison of the information content of the two methods, complementary to the DESI-like analysis.

3 Environment estimators

Both estimators are evaluated on the *full periodic box*, and on the same set of $N_r = N_g$ uniform random points (fixed seed, drawn in the unit cube and rescaled to the box, so that the random points are identical in all cosmologies up to the AP rescaling).

3.1 ASTRA split

The galaxies and random points are joined into a single catalogue and Delaunay-triangulated. For every point (galaxy or random), with $N_{\rm data}$ and $N_{\rm rand}$ the numbers of galaxies and randoms among its Delaunay neighbours,

$$r = \frac{N_{\rm data} - N_{\rm rand}}{N_{\rm data} + N_{\rm rand}} \in [-1, 1]. \qquad (1)$$

A point surrounded by galaxies ($r \to 1$) lies in a dense region, a point surrounded by randoms ($r \to -1$) in an underdense one. In the original ASTRA classification [4], thresholds on r define voids, sheets, filaments and knots; here we use r as a continuous environment variable. The triangulation is computed independently in each subvolume (Section 4) on a region padded by $30\,h^{-1}$Mpc on each side, with periodic wrapping across the faces of the box, and r is kept only for the points inside the subvolume. This removes the edge effects of a non-periodic triangulation (missing neighbours, elongated boundary simplices). The mean inter-point separation of the joint catalogue is $\simeq 10\,h^{-1}$Mpc, so the padding contains several layers of neighbours (convergence in Section 8).

3.2 Density split

As a reference we use the standard density split [1]. The galaxies are assigned to a periodic mesh of $5\,h^{-1}$Mpc cells with a cloud-in-cell (CIC) scheme, the overdensity field is smoothed with a Gaussian kernel of radius $R_s = 10\,h^{-1}$Mpc by FFT, and the smoothed density contrast δ_R is read (CIC) at the position of every galaxy and random point.

3.3 Environment quartiles

For each method, each subvolume and each population, points are ranked by their environment value (r or δ_R) and split into four quartiles of exactly equal size, Q1 (most underdense) to Q4 (most overdense). Because r is a ratio of small integers, many points share the same value; ties are broken at random. We consider two populations, ranked separately so that the thresholds of one do not depend on the other:

- **random quartiles**: the random points ranked by their environment, the query points of density split, sampling environments by volume;
- **galaxy quartiles**: the galaxies ranked by their own environment, sampling environments by mass.

3.4 Cosmic-web classes and why we use quartiles

ASTRA was designed to classify points into four cosmic-web classes with fixed thresholds on r [4]: voids ($r \le -0.9$), sheets ($-0.9 < r \le 0$), filaments ($0 < r \le 0.9$) and knots ($r > 0.9$). We use exactly the definition of the reference implementation (Eq. 1 with $N_r = N_g$, r from a single random catalogue, these thresholds). On the DESI-like fiducial catalogue the classes are extremely unbalanced (Figure 3): 0.25% of the random points are voids, 65% sheets, 34% filaments and 0.05% knots; for the galaxies 0.07%, 49%, 51% and 0.3%. In Quijote, 0.7%, 72%, 27% and 0% of the random points. In practice the classification reduces to *two dominant classes*, sheets and filaments, i.e. to a split of r at zero; the voids and knots contain a few hundred points per subvolume, too few for precise clustering measurements (Figure 1). This imbalance motivated the use of equal-size quartiles of r in the main analysis, which keep the ranking of the environments by ASTRA while giving every environment the same statistical weight, as in density split.

The ASTRA paper reports 7–11% of random points in voids [4], against 0.25–0.7% here. Its samples are 11–73 times denser than our LRGs (number densities 1.7×10^{-3} to $1.1 \times 10^{-2}\,{\rm Mpc}^{-3}$, against $1.5 \times 10^{-4}\,{\rm Mpc}^{-3}$). The fractions are not set by the density alone (the Quijote halos, ten times sparser than the LRGs but more strongly clustered, have more voids), but by the density contrast of the tracer

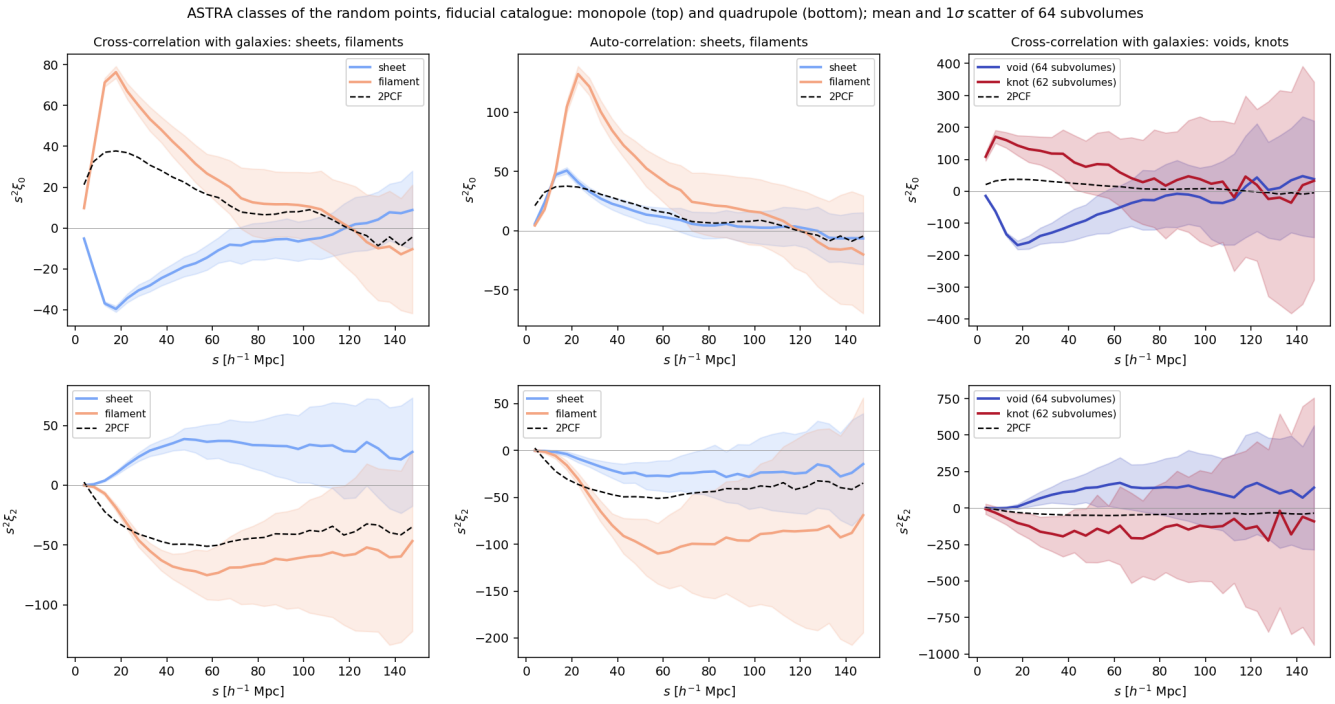


Figure 1: Monopole (top) and quadrupole (bottom) of the ASTRA classes of the random points on the fiducial catalogue: cross-correlation with the galaxies (left) and auto-correlation (centre) of the sheets and filaments, and cross-correlation of the voids and knots (right, different scale). Lines and bands: mean and 1σ scatter of the 64 subvolumes; dashed: 2PCF. The filaments behave like the overdense quartile Q4 and the sheets like the underdense quartile Q1; the voids and knots contain only ~ 150 and ~ 30 random points per subvolume, hence their large scatter.

field at the scale of the mean inter-point separation. This is another manifestation of the adaptive scale of ASTRA (Section 8): fixed thresholds on r do not select the same environments in samples of different density and bias, whereas quantiles of r do define comparable environments.

3.5 Comparison of the two classifications

Figure 2 compares the two estimators on the fiducial catalogue. They are strongly but not perfectly correlated: the Spearman rank correlation between r and δ_R is 0.719 ± 0.005 for random points and 0.713 ± 0.005 for galaxies (scatter over subvolumes). The extreme quartiles are the most consistent: 63% of the ASTRA Q1 (Q4) random points are also in the density-split Q1 (Q4), whereas the intermediate quartiles are largely mixed, and less than 2% of the points move between the two extremes. ASTRA identifies the same large-scale structures as density split but redistributes many points (Figure 3), as expected from an adaptive rather than a fixed scale. The distributions of r and δ_R per quartile are shown in Figure 4.

Response to cosmology. Figure 5 shows how the mean environment of each quartile of the Quijote halos shifts when a cosmological parameter is varied, in units of its scatter between realisations. The density-split contrast δ_R responds strongly (up to 7σ for σ_8), antisymmetrically for the $\pm$ variations; the ASTRA value r hardly moves ($< 1\sigma$). This is expected: r is normalised locally by the random points and is nearly insensitive to the overall amplitude of the field. The cosmological information of ASTRA split therefore lies in the clustering of its environments rather than in the values of r.

4 Clustering measurements

4.1 Subvolumes and estimator

Each $2\,h^{-1}$Gpc box is divided into $4^3 = 64$ non-overlapping cubic subvolumes of $500\,h^{-1}$Mpc (about 62 500 galaxies each). Each subvolume is treated as an independent survey with open boundaries: correlation functions are measured with the Landy–Szalay estimator [10],

$$\xi(s,\mu) = \frac{D_1D_2 - D_1R - RD_2 + RR}{RR}, \qquad (2)$$

using a uniform catalogue of $5\times$ the number of galaxies to describe the geometry (the same for all subvolumes, so that RR is computed once). Pair counts are computed with `pycorr/Corrfunc` [11] in 30 bins of $5\,h^{-1}$Mpc in $s \in [0, 150]\,h^{-1}$Mpc and 120 bins in

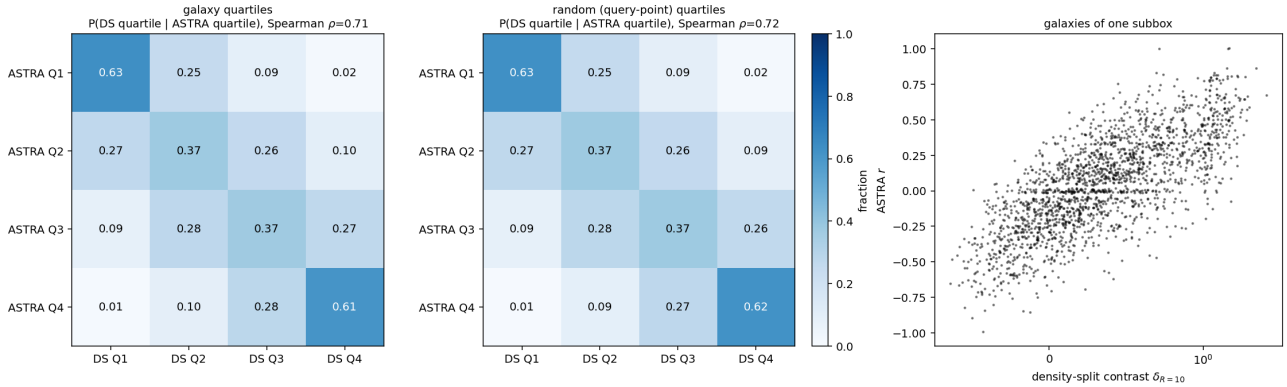


Figure 2: ASTRA and density-split environments on the fiducial catalogue (`c000`, `hod239`, line of sight z). Left, middle: probability that a point of a given ASTRA quartile falls in each density-split quartile, for galaxies and random points (average over 64 subvolumes). Right: ASTRA r against the smoothed density contrast for a sample of galaxies (small vertical jitter added to the discrete r).

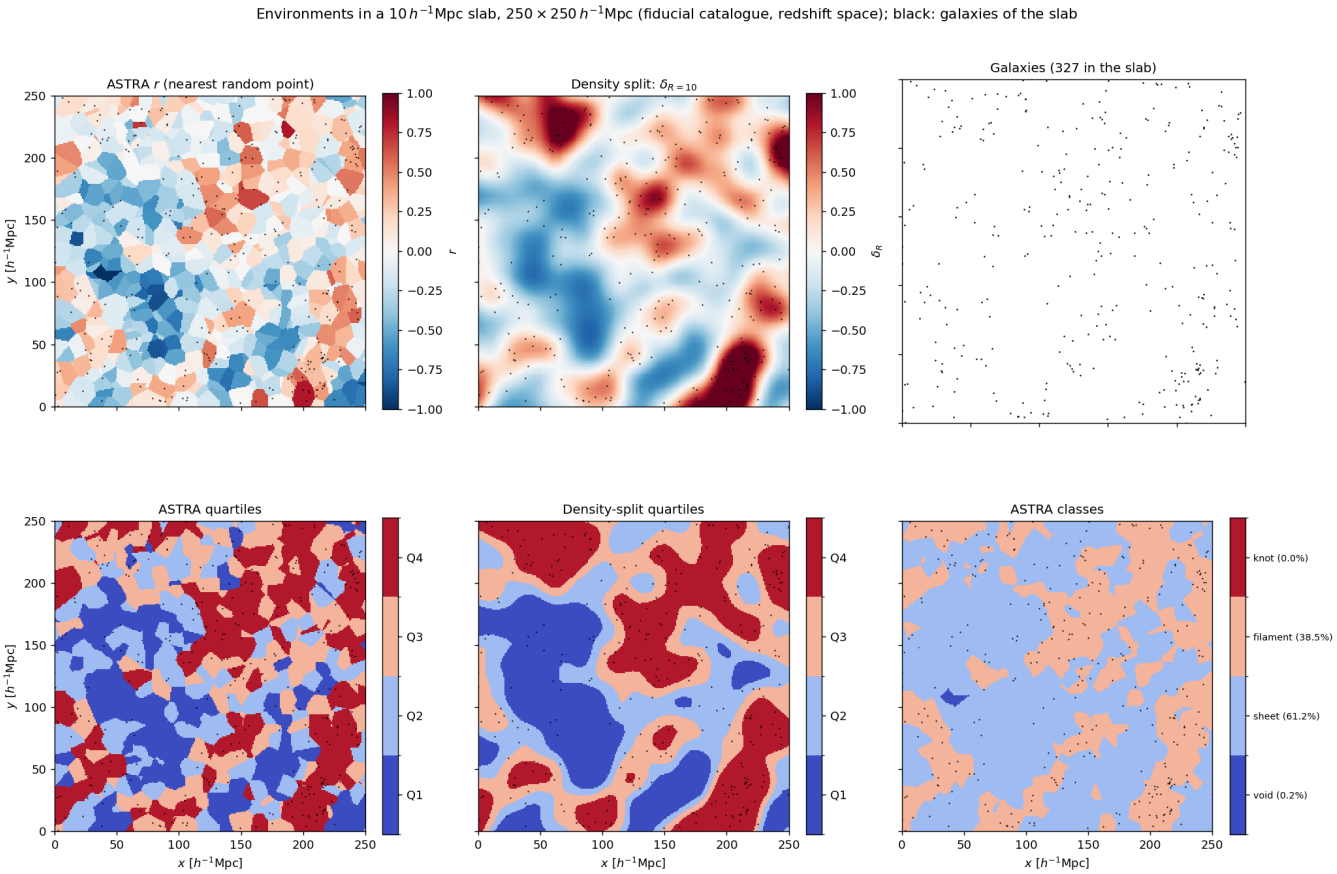


Figure 3: Environments in a $10\,h^{-1}$Mpc slab of $250 \times 250\,h^{-1}$Mpc of the fiducial catalogue, drawn as filled regions: each pixel takes the ASTRA value r, quartile or class of the nearest random point (left, bottom right), or the smoothed density contrast and its quartile (centre); black dots: galaxies of the slab. ASTRA defines environments cell by cell, at the scale of the Delaunay neighbours, whereas density split defines them on a smooth field; the ASTRA classes are dominated by sheets and filaments.

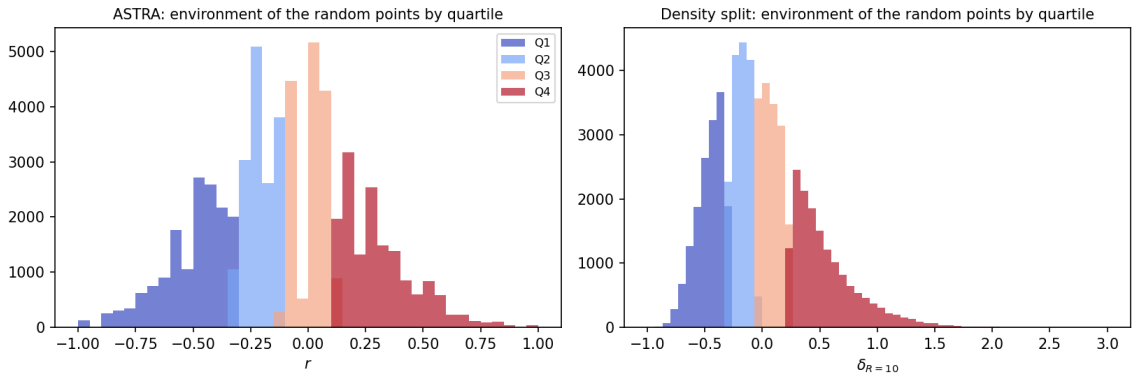


Figure 4: Distribution of the environment of the random points per quartile: ASTRA r (left, discrete values) and density-split δ_R (right).

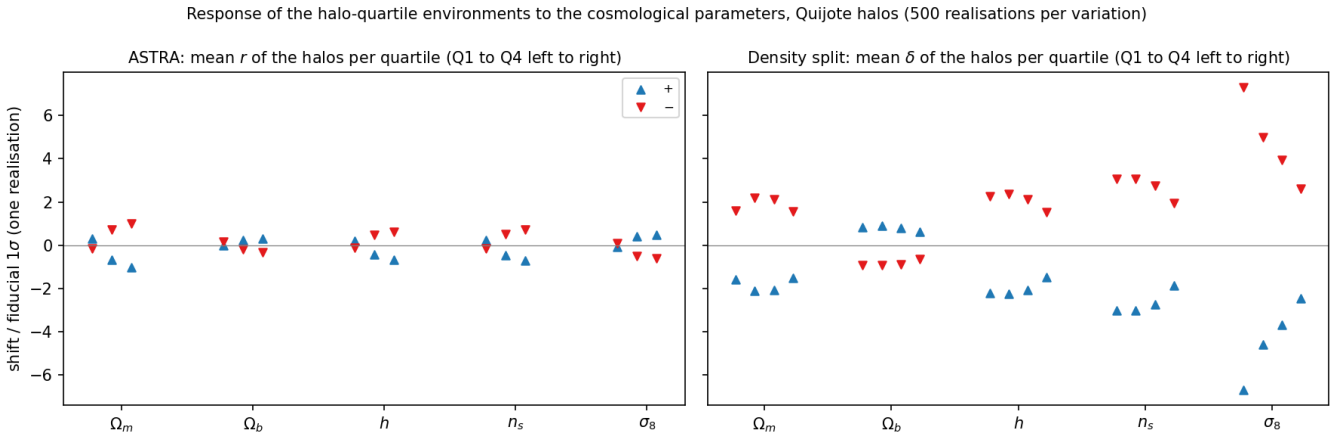


Figure 5: Shift of the mean environment of each halo quartile (Q1 to Q4 from left to right) for the + (blue) and − (red) variations of each parameter, Quijote, in units of the scatter of one realisation. Left: ASTRA r; right: density-split δ_R.

$\mu \in [-1, 1]$, with μ defined with respect to the fixed line of sight, and projected on the Legendre multipoles $\ell = 0, 2, 4$ (only $\ell = 0, 2$ are used in the forecasts).

4.2 Statistics

For each method $m \in \{\text{ASTRA, DS}\}$, population $p \in \{\text{random, galaxy}\}$ and quartile Q_i we measure the quartile–galaxy cross-correlation $\xi^{\text{QG}}_{i,\ell}(s)$, between the points of Q_i and all galaxies, and the quartile auto-correlation $\xi^{\text{QQ}}_{i,\ell}(s)$, together with the galaxy two-point correlation function $\xi^{\text{GG}}_{\ell}(s)$ (2PCF): 33 statistics per catalogue. For random quartiles, ξ^{QG} and ξ^{QQ} are the analogues of the density-split cross- and auto-correlation functions [1]. For galaxy quartiles, which partition the galaxies into four equal sets, the average of the four quartile–galaxy cross-correlations is the 2PCF: the quartiles decompose the 2PCF by environment.

4.3 Results on the fiducial cosmology

Figure 6 compares the extreme quartiles of the two methods, and Figures 7 and 8 show all four quartiles of each method in the layout of the density-split analyses [1]; the galaxy quartiles are shown in Figure 9.

The two methods produce qualitatively identical signals. The Q1–galaxy cross-correlation is negative at small separations, the profile of galaxies around underdense regions, and the Q4–galaxy cross-correlation is positive and above the galaxy 2PCF, the profile around overdense regions. The auto-correlations of the extreme quartiles are strongly enhanced with respect to the 2PCF. The quadrupoles have opposite signs for Q1 and Q4: around underdense regions, peculiar velocities point outwards and stretch the void profile along the line of sight, while around overdense regions infall compresses it. The intermediate quartiles Q2 and Q3 have weak cross-correlations of opposite signs, close to the mean field, and auto-correlations 5–6 times below those of the extreme quartiles.

The ASTRA signals are systematically weaker, by a factor ~ 1.6 at the peak of the Q1 and Q4 cross-

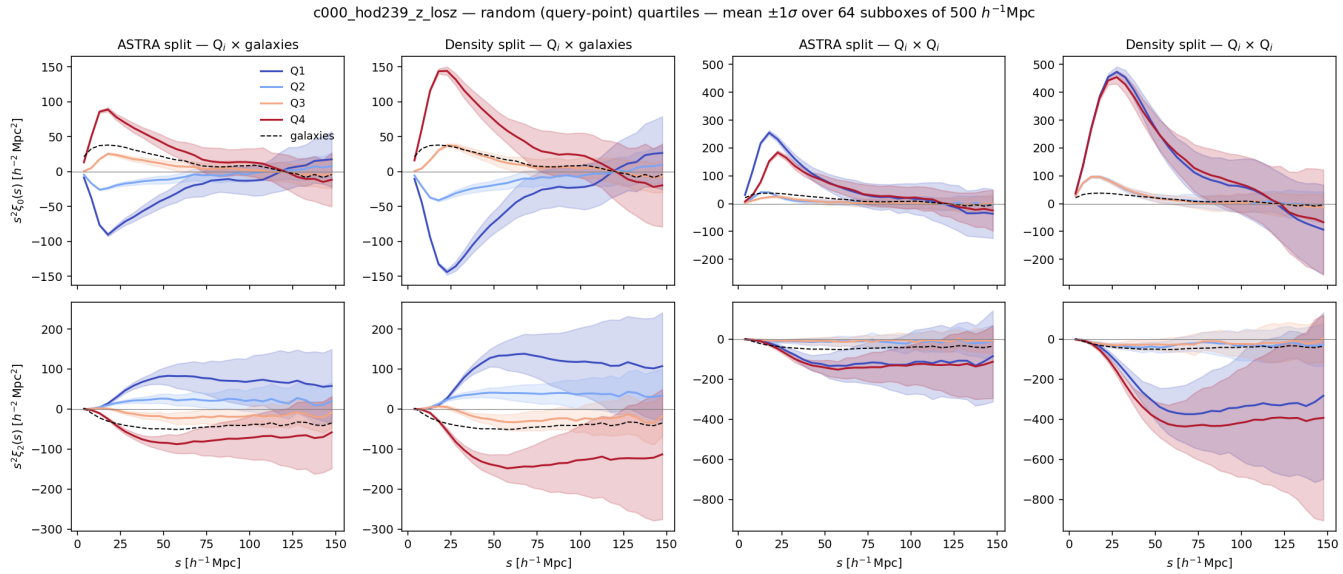


Figure 6: Monopole (top) and quadrupole (bottom) of the random-quartile statistics on the fiducial catalogue, for ASTRA split and density split: quartile–galaxy cross-correlations (left two columns) and quartile auto-correlations (right two columns), from Q1 (blue) to Q4 (red); dashed: galaxy 2PCF. Mean and 1σ scatter over the 64 subvolumes of $(500\,h^{-1}\mathrm{Mpc})^3$; panels of the same statistic share the same scale.

correlations ($s^2\xi_0 \simeq -90$ and $+90\,h^{-2}\mathrm{Mpc}^2$, against -145 and $+145$ for density split). This is consistent with the partial mixing of the quartiles: ASTRA relies on $\sim$ 15 neighbours per point, which dilutes the contrast between environments. For density split the Q1 and Q4 auto-correlations are nearly equal, whereas for ASTRA the Q1 auto-correlation peaks 40% above the Q4 one: ASTRA's underdense quartile is relatively more clustered with itself, a first sign that its auto-correlations carry distinct information. The relevant question is not the amplitude of the signal but its information content relative to its noise, which we address with the forecasts.

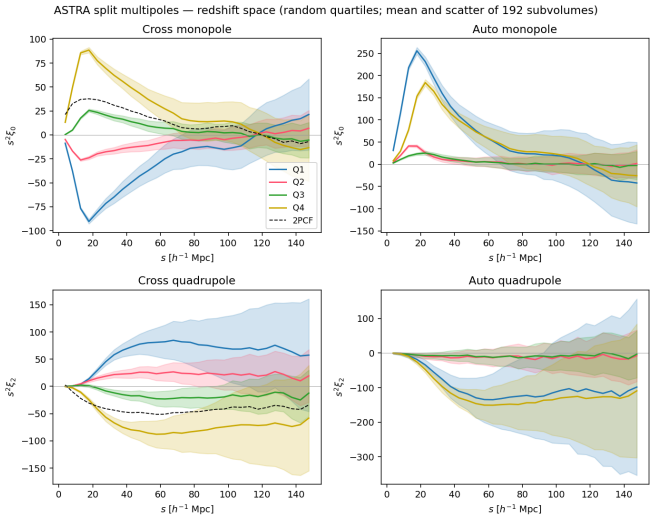


Figure 7: ASTRA split: monopole and quadrupole of the quartile–galaxy cross-correlations (left) and quartile auto-correlations (right) for the four random quartiles; dashed: galaxy 2PCF. Mean and scatter of 192 subvolumes of the fiducial catalogue.

4.4 Real space and redshift space

To isolate the effect of RSD, we also measured the fiducial catalogue (and the eight catalogues of the derivative pairs) in real space, the environments being then defined from the real-space positions (r-split) instead of the redshift-space ones (z-split), along the three lines of sight. Figure 10 compares the two. RSD increase the monopoles at 20–60 h^{-1}Mpc by a factor 1.6 for the galaxy 2PCF and by 1.3–1.5 for the Q1 and Q4 cross-correlations of both methods, and create quadrupoles that are absent in real space: positive around underdense regions (outflows) and negative around overdense ones (infall). In absolute value the quadrupoles of density split are 1.6–1.7 times larger than those of ASTRA split, following their larger monopoles; relative to the monopole, however, the ASTRA quadrupole is the larger one ($|\xi_2|/|\xi_0| \simeq 0.8$ for ASTRA vs 0.6 for density split for Q1, 1.0 vs 0.8 for Q4, at 20–60 h^{-1}Mpc): the ASTRA environments are less contrasted but their redshift-space anisotropy is relatively stronger. Real-space environments are not observable, so all the forecasts use z-split; the real-space measurements are kept as a diagnostic. (A first version of this work used real space as a "truth" to quantify "systematic biases" of redshift-space forecasts; since both measurements and emulator are in redshift space, this comparison does not describe a bias of the analysis, and the relevant systematic, the emulator error, is included in

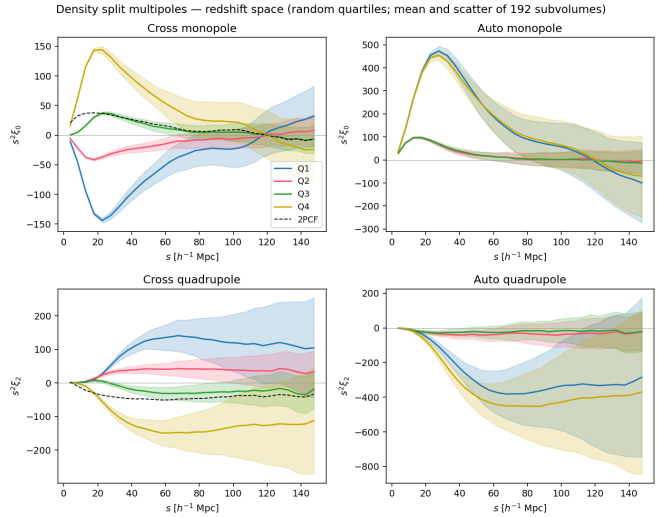


Figure 8: Same as Figure 7 for density split.

the covariance.)

5 Covariance, derivatives and Fisher forecasts

Data vectors. By default, the cross-correlations of the extreme quartiles Q1 and Q4 with the galaxies, $\ell = 0, 2$, $20 \le s \le 150\,h^{-1}$Mpc in $10\,h^{-1}$Mpc bins (13 bins per multipole, 52 numbers per method), with or without the 2PCF (26 numbers). Variants add the quartile auto-correlations, use the galaxy quartiles, single quartiles, or change the minimum scale.

5.1 Covariance

Baseline: 1786 independent boxes. The covariance is measured on the 1786 official small boxes (Section 2.2): 1786 independent $(500\,h^{-1}\mathrm{Mpc})^3$ volumes treated exactly like our subvolumes. Forecasts are quoted for this volume of $0.125\,(h^{-1}\mathrm{Gpc})^3$. The correlation matrix (Figure 11) shows strong correlations between neighbouring scales, between the Q4 cross-correlations and the 2PCF, and anti-correlations between the Q1 and Q4 blocks. The ASTRA and density-split blocks of the same quartile are very strongly correlated: being measured on the same galaxies, their noise is largely common. This is the origin of their complementarity: when two measurements share most of their noise but respond differently to the parameters, their combination cancels part of that noise and gains much more than independent measurements would.

First approach: 192 subvolumes of one box. Before the official mocks were used, the covariance was estimated from the fiducial catalogue of the emulator suite (`c000`, `hod239`, the c000 draw closest to the centre of the HOD prior), using the 64 subvolumes measured along the three lines of sight, $N_s = 192$ samples. The three lines of sight are not independent, an approximation commonly used in simulation-based analyses. The inverse covariance is corrected by the Hartlap factor [14], $(N_s - N_d - 2)/(N_s - 1)$: 0.72 for one method ($N_d = 52$), 0.59 with the 2PCF ($N_d = 78$), 0.31 for ASTRA + DS + 2PCF ($N_d = 130$), which motivated the compression of the data vectors (Section 5.5). Section 6 shows how replacing this covariance by the independent one changed the comparison.

5.2 Finite differences: the HOD-mismatch problem

The AbacusSummit derivative grid provides, for each parameter, two simulations differing only by a small change of that parameter and sharing the same initial conditions. A natural estimate of the derivatives is the paired finite difference: for each parameter we select one HOD file in each cosmology of the pair, minimising the distance between the two HOD parameter vectors (Table 3), and difference the statistics subvolume by subvolume and line of sight by line of sight before averaging, so that the cosmic variance common to the two simulations largely cancels. This cancellation is only partial: each HOD catalogue is an independent draw of the galaxies populating the halos, so the galaxy sampling noise is not shared, and the difference of two full-box measurements with different HODs has an error comparable to that of a single box (1.05 times, median over the pairs of the grid). The precision of the paired derivatives comes from the average over the 192 paired samples.

Parameter	HOD files	$\lvert\Delta h\rvert$	random pairs
fiducial	c000_hod239	–	–
ω_b	c100_hod448 / c101_hod694	0.63	1.38
ω_{cdm}	c102_hod589 / c103_hod841	0.58	1.38
n_s	c104_hod308 / c105_hod735	0.65	1.37
σ_8	c112_hod230 / c113_hod204	0.65	1.39

Table 3: HOD catalogues of the finite-difference pairs. $|\Delta h|$ is the distance between the two HOD parameter vectors normalised to the prior range (12 parameters); the last column gives the median for random pairs.

Averaged over the 192 samples, these differences have small statistical errors, but they are not cosmological derivatives: because the HODs of the two cosmologies differ, each difference is dominated by the response to the HOD mismatch, 10–15 times larger than the response to the cosmological parame-

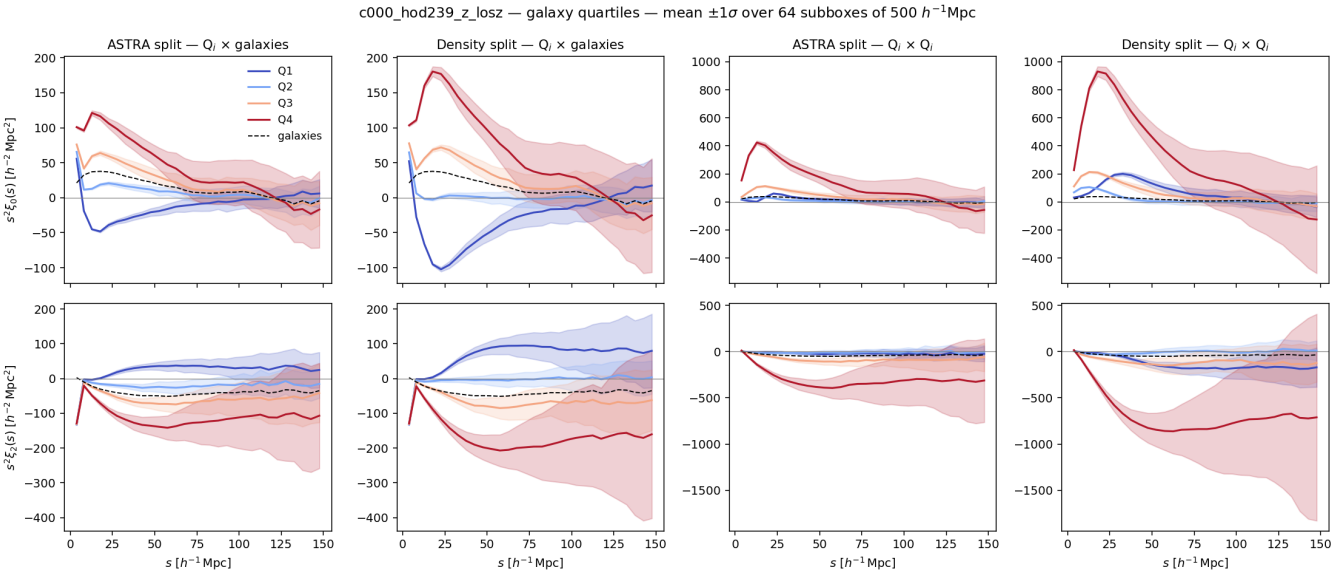


Figure 9: Same as Figure 6 for the galaxy quartiles.

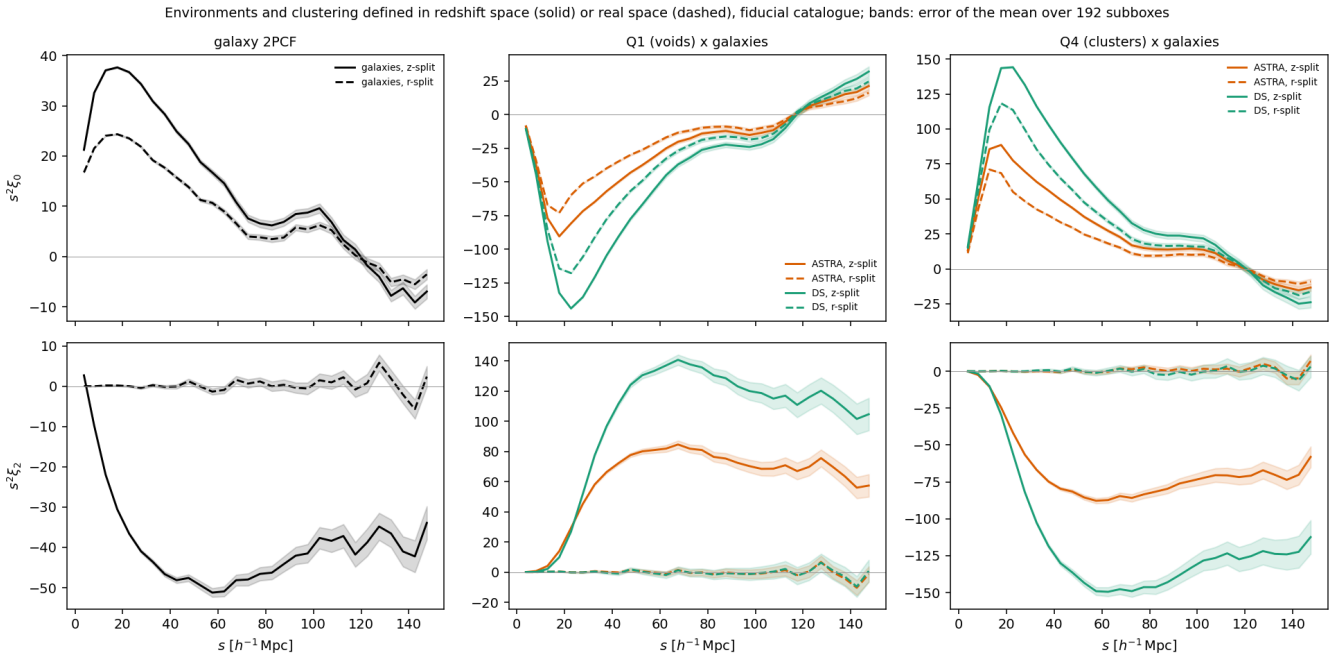


Figure 10: Environments and clustering defined in redshift space (solid) or real space (dashed) for the fiducial catalogue: galaxy 2PCF (left), Q1 (centre) and Q4 (right) random-quartile cross-correlations of ASTRA split (orange) and density split (green). Top: monopole, bottom: quadrupole. Bands: error of the mean over 3×64 subvolumes.

ter. The derivative vectors with respect to ω_b, n_s and σ_8 are almost collinear (correlations 0.95–0.99 in the Fisher metric) and the Fisher matrices are not positive definite. We tried to calibrate and subtract the HOD response with 72 additional fiducial catalogues spanning the HOD parameters of the pairs: a linear model predicts left-out catalogues with a 45% error, a Gaussian-process model (one GP per principal component of the data vector, anisotropic kernel) with 8%. Since the HOD term is an order of magnitude larger than the cosmological one, even 8% leaves a residual comparable to the signal, and the corrected Fisher matrices remain unstable.

The finite differences are nevertheless not discarded. They (i) diagnose and quantify the limitation of an emulator suite for derivatives, a pitfall generic to suites in which the HOD is sampled independently in each cosmology; (ii) once corrected for the HOD response, provide a check of the emulator derivatives that does not rely on a neural network (Figure 14); (iii) with several HOD draws per cosmology, become an independent estimator of the derivatives by re-

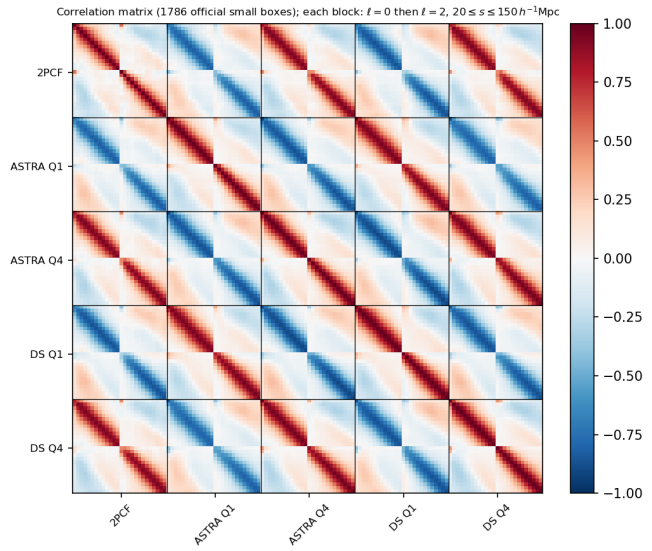


Figure 11: Correlation matrix of the 2PCF and of the Q1, Q4 cross-correlations of ASTRA split and density split (1786 official small boxes).

gression over all the pairs (Section 5.4).

5.3 Neural-network emulator

Why an emulator. As in SUNBIRD [3], we model each statistic as a smooth function of the cosmological and HOD parameters, and take its derivatives at exactly fixed HOD by automatic differentiation, which removes the HOD mismatch by construction.

Inputs and outputs. The inputs are the 8 cosmological parameters varied in AbacusSummit (ω_b, $\omega_{\rm cdm}$, σ_8, n_s, α_s, $N_{\rm ur}$, w_0, w_a; h and A_s follow from θ_* and σ_8) and the 12 HOD parameters, rescaled to $[0,1]$ over the prior. The output of one network is one statistic, the monopole and quadrupole for $20 \le s \le 150\,h^{-1}$Mpc in $10\,h^{-1}$Mpc bins (26 numbers), measured on the full box as the mean of the 64 subvolumes, standardised by its mean and by the statistical error of the full box $\sigma_{\rm box}$. There is one network per statistic (33 statistics).

Architecture and training. Fully-connected networks with three hidden layers of 256 units and SiLU activations, smooth so that their derivatives are well behaved, trained with Adam and early stopping on a validation set. With the standardisation above the loss is a χ^2 per bin. Each statistic has an ensemble of 10 networks with different initialisations; the prediction is the ensemble mean, and the scatter between members measures the uncertainty due to training.

Training set. 999 catalogues of the 85 cosmologies: 8 HOD draws per cosmology, the 72 fiducial calibration catalogues, and 16 draws close to the fiducial HOD in c000 and in each of the 14 cosmologies of the derivative grid, where the derivatives are evaluated (densification, below). Validation and test sets are 8 cosmologies each, never seen in training.

Accuracy. On the test cosmologies the emulator reproduces all 33 statistics with a median error of 0.4–0.9 $\sigma_{\rm box}$, about ten times below the error of the $(500\,h^{-1}{\rm Mpc})^3$ volume of the forecasts (Figure 12); the emulator error is nevertheless added to the covariance. A learning curve, retraining on 25, 50, 75 and 100% of the training set, shows that this error has reached a plateau at $\simeq 0.5\sigma_{\rm box}$, set by the galaxy sampling noise of single catalogues: more catalogues would barely help (Appendix C).

Cosmological response. Accurate predictions do not guarantee accurate derivatives. We test the latter by training the emulator without the 14 cosmologies of the derivative grid and comparing its prediction of the $\pm$ pair differences, HOD mismatch included, to the simulations (Figure 13). The response to σ_8 and n_s is reproduced to 10–20%, the response to ω_b to 30–45% with the correct shape, but the response to $\omega_{\rm cdm}$ is not captured below $s \simeq 70\,h^{-1}$Mpc: at fixed θ_* and σ_8 its effect is smaller than the emulator error and is not learnt away from the training points. This led us to *densify* the training set around the fiducial point with the 16 HOD draws closest to the fiducial HOD in c000 and in each cosmology of the grid (240 catalogues), so that the emulator sees the response to each parameter for many similar galaxy–halo connections. Densification alone did not stabilise the derivatives: the Fisher errors of the individual networks scattered by 70–170% for the 2PCF and for $\omega_{\rm cdm}$ and n_s, and the ASTRA/density-split comparison changed sign between trainings. As an independent check, the derivatives of the emulator at fixed HOD agree with the GP-corrected pair differences to 1–3% for n_s and $\omega_{\rm cdm}$, 5–6% for σ_8 and 8–13% for ω_b (Figure 14).

Training on catalogue differences. To constrain the response near the fiducial point directly, we add to the loss the χ^2 of the predicted *differences* between catalogues, over 2544 pairs: all the pairs $(c_+, h_i), (c_-, h_j)$ of the seven $\pm$ pairs of the grid with HODs within one prior range of the fiducial, and pairs of the same cosmology (HOD response); details in Appendix B. The differences are not much more precise than individual measurements, since the galaxy sampling noise is not shared, but the term constrains their *structure*: the network must reproduce how the statistics change between nearby points of parameter

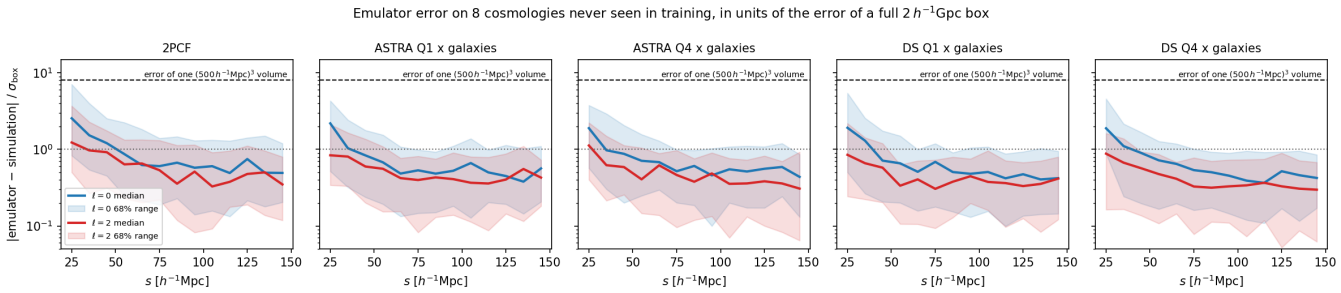


Figure 12: Emulator accuracy on 8 held-out cosmologies (8 HOD draws each): median (lines) and 68% range (bands) of |emulator − simulation| in units of the statistical error of a 2 h^{-1}Gpc box, for the monopole (blue) and quadrupole (red) of the 2PCF and of the Q1, Q4 cross-correlations of ASTRA split and density split. Dashed: error of one (500 h^{-1}Mpc)^3 volume, the volume of the forecasts.

space, which is what the derivatives require. We also tested an additional weight $(25\,h^{-1}\mathrm{Mpc}/s)^\alpha$, $\alpha = 1$, on the small scales. We validated the three variants on 56 local catalogues (4 per grid cosmology) excluded from training, by predicting the differences of all the held-out ± pairs (Table 4, Figure 15). The difference term halves the error for ω_b and σ_8 and reduces it for all parameters but $\omega_{\rm cdm}$, already reproduced to 11% near the fiducial; the residuals reach the noise level of the measured differences ($\simeq 1\sigma_{\rm box}$), so the data do not allow a more stringent test. The small-scale weight brings a marginal additional gain but makes the forecasts that include the quartile auto-correlations unstable, so we use the difference term without it: this is the baseline emulator.

	ω_b	$\omega_{\rm cdm}$	n_s	α_s	w_0	w_0, w_a	σ_8
no diff. term	21	11	35	14	17	17	13
diff. term	10	11	27	10	12	10	7
+ small scales	8	11	26	9	11	8	7

Table 4: Median relative error (%) of the emulator prediction of the differences between held-out local catalogues of the ± pairs of the derivative grid, for the three training variants.

Derivatives. Figure 16 shows the baseline derivatives of the 2PCF and of the Q1, Q4 cross-correlations in units of the statistical error: the two methods respond very similarly to $\omega_{\rm cdm}$ and n_s, mostly through the monopole at $s < 75\,h^{-1}$Mpc; the Q1 (void) cross-correlations respond with the opposite sign to Q4 and to the 2PCF, which is what breaks degeneracies; σ_8 acts mostly on the quadrupoles at fixed HOD. Figures 17 and 18 show the derivatives of the four quartiles in the layout of the first version: the response is ordered from Q1 to Q4 for $\omega_{\rm cdm}$ and n_s, the extreme quartiles responding most; for σ_8 at fixed HOD the monopole responses are below 1σ for density split and the information comes from the quadrupoles.

5.4 Derivatives from a regression over pairs

As an alternative to the emulator, we estimate the derivatives locally from the derivative grid alone. For a pair of cosmologies $c_\pm$, all the catalogues of each cosmology within a distance $r_{\rm max}$ of the fiducial HOD h_0 (the 16 draws of the densification, the pair draw and the emulator draws) are fitted with a first-order expansion in the HOD parameters, $\boldsymbol{d}_c(h) = \boldsymbol{a}_c + G\,(h - h_0)$, with a response G shared by the two cosmologies; the derivative is $(\boldsymbol{a}_{c_+} - \boldsymbol{a}_{c_-})/(\theta_+ - \theta_-)$, the difference between the two cosmologies at exactly the fiducial HOD. The fit is performed subvolume by subvolume, so that part of the cosmic variance cancels, and the errors are obtained by bootstrapping the subvolumes. The intercepts are interpolated in the middle of draws surrounding h_0, and curvature common to the two cosmologies cancels in the difference. This estimator relies on a linear model and on the derivative grid only, while the emulator is a non-linear model of the whole suite: agreement between the two is a strong test of the derivatives.

Reduced HOD space. With 12 HOD parameters and ∼ 18 catalogues per cosmology, the fit is poorly conditioned: its leave-one-out error is 0.8–1.0 times the scatter between HODs, and separate responses $G_\pm$ or quadratic terms do not help. We therefore restrict it to the most important directions of the HOD space, determined independently from the 72 fiducial calibration catalogues: the right singular vectors of the linear HOD response, each weighted by its statistical error. The singular values are $1, 0.054, 0.014, 0.013, \ldots$: *the clustering statistics respond essentially to a single combination of the 12 HOD parameters*, plausibly the large-scale galaxy bias. Keeping 3 or 4 directions, the leave-one-out error drops to 0.7–0.9 times the HOD scatter, at the

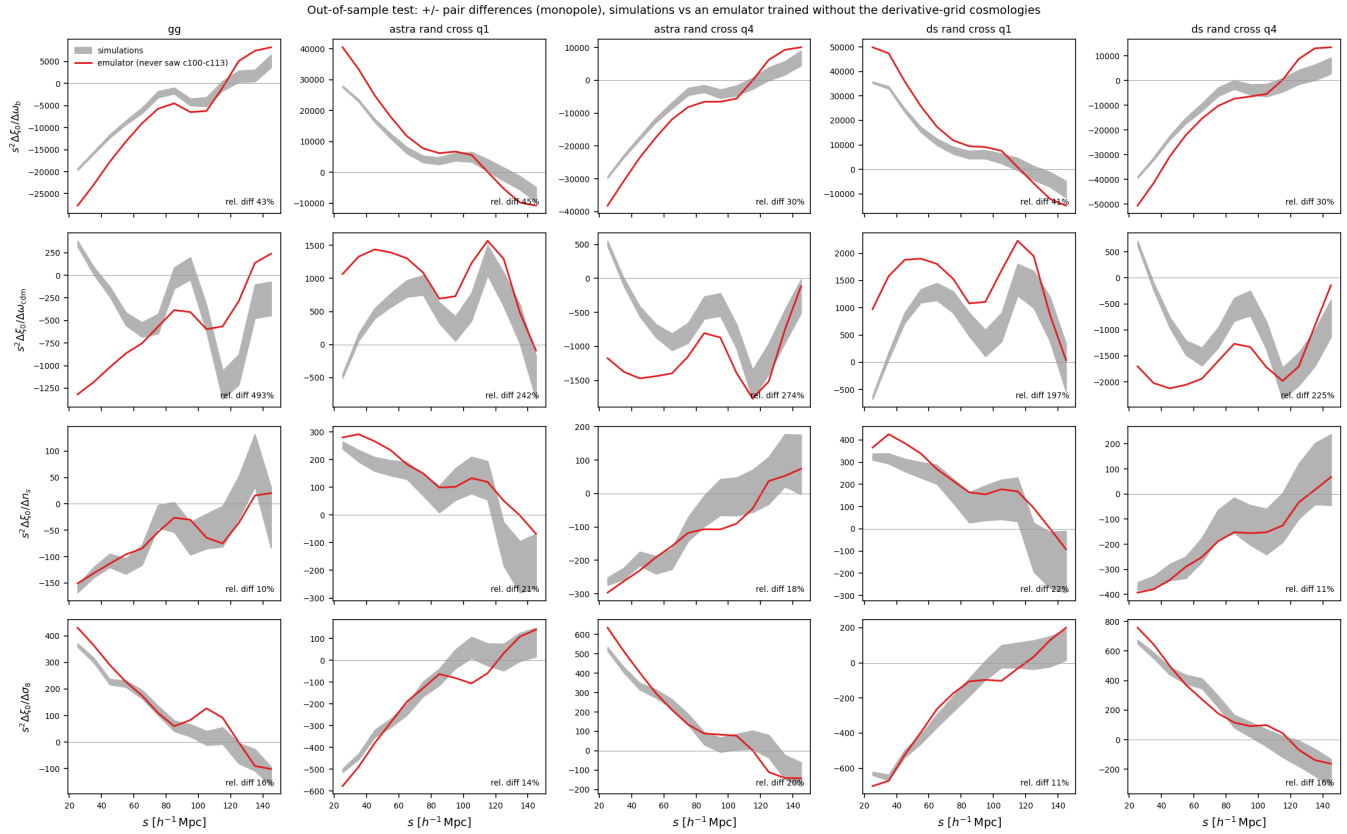


Figure 13: Out-of-sample test of the cosmological response: monopole of the ± pair differences in the simulations (grey band: mean ± error over the 192 paired samples) and predicted by an emulator trained without the cosmologies of the derivative grid (red).

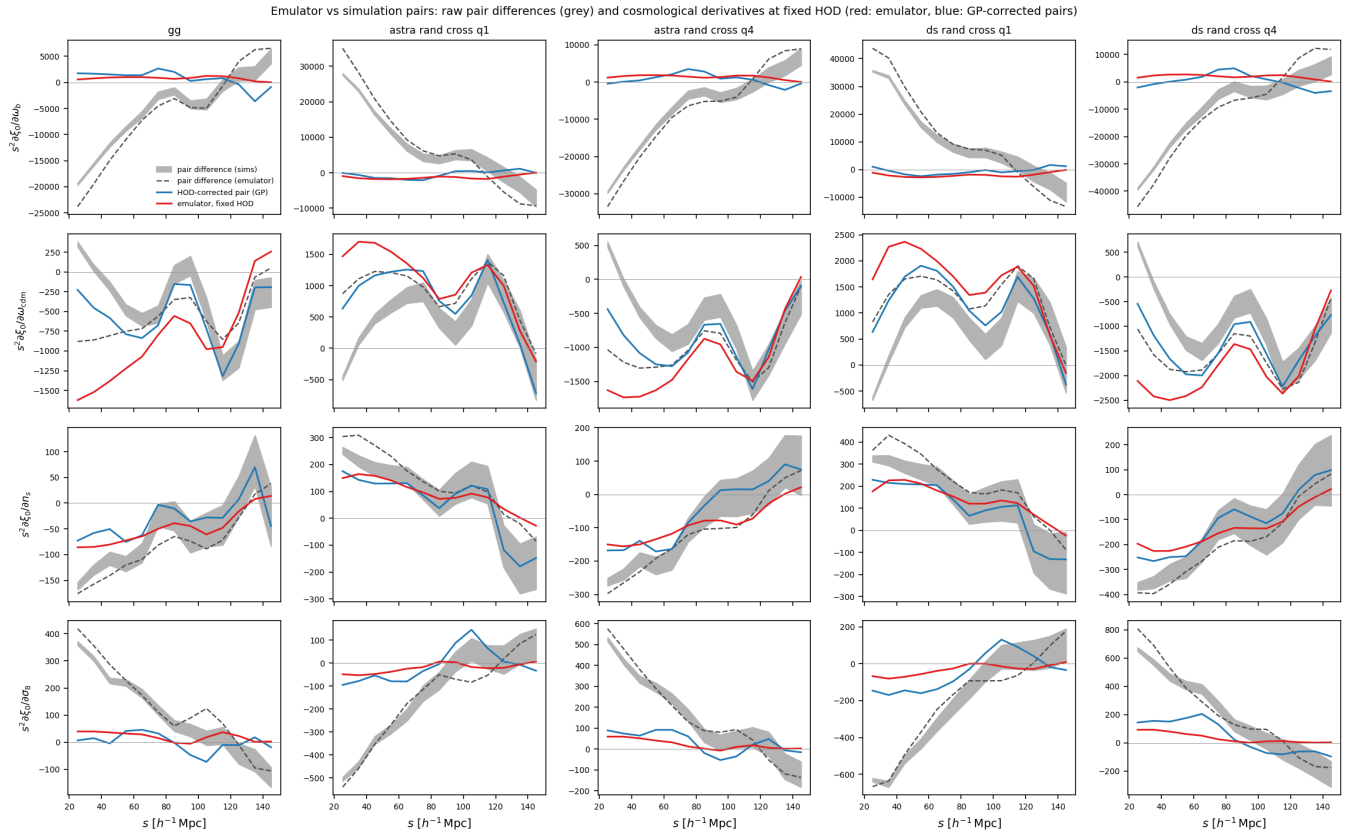


Figure 14: Cosmological derivatives at fixed HOD from the emulator (red) and from the pair differences corrected with the GP calibration of the HOD response (blue), with the raw pair differences in the simulations (grey) and as predicted by the emulator (dashed).

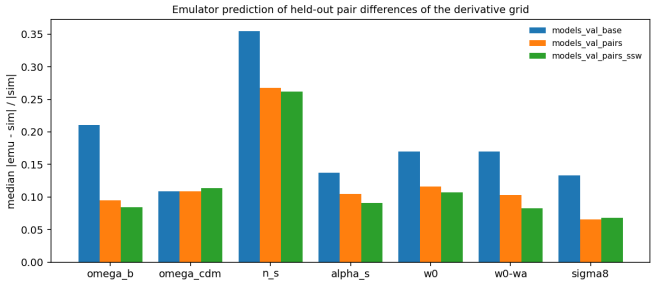


Figure 15: Median relative error of the emulator prediction of held-out pair differences of the derivative grid, for the three training variants (Table 4).

level of the measurement noise, and the two choices give the same forecasts within 5%. We use 3 directions.

5.5 Fisher matrices

The derivatives $D = \partial \boldsymbol{d}/\partial(\boldsymbol{\theta}, \boldsymbol{h})$ are obtained at the fiducial point for the 6 cosmological parameters $\boldsymbol{\theta} = (\omega_b, \omega_{\rm cdm}, n_s, \sigma_8, w_0, w_a)$ and the 12 HOD parameters $\boldsymbol{h}$. The covariance is the sum of the measured covariance and of the (diagonal) emulator error. The Fisher matrix $F = D^T \hat{C}^{-1} D + P_h$ includes a Gaussian prior P_h on the HOD parameters of width 0.29 times the prior range (the standard deviation of the uniform prior); the cosmological errors are marginalised over the HOD by inverting the full 18×18 matrix, and we also quote errors at fixed HOD. Marginalising over the HOD degrades the constraints by less than $\sim 30\%$, the most affected parameter being n_s. The uncertainty due to the emulator training is estimated from the scatter of the forecasts of the individual networks, divided by $\sqrt{10}$.

Compression. Each data vector is compressed to one number per parameter (18 numbers), $\boldsymbol{t} = W^T \boldsymbol{d}$, $W = \tilde{C}^{-1} D$ ($\tilde{C}$: Ledoit–Wolf shrinkage estimate of the covariance), whose covariance is then estimated from the samples (Hartlap factor 0.91 with 192 samples, 0.99 with 1786). The compression is lossless if $\tilde{C} = C$ and otherwise loses information without biasing the forecast; it allows long and combined data vectors to be compared fairly.

6 Analysis choices and how the results evolved

The framework above is the result of successive corrections, each motivated by a failure of the previous approach (Table 5, details in Appendix C). We report the intermediate results because they show which conclusions depend on the analysis choices.

Results with the single-box covariance. With the 192-subvolume covariance and the `hod239` fiducial, the baseline emulator and the regression over pairs, which rely on different assumptions, agreed that ASTRA split was as constraining as or slightly more constraining than density split for the random quartiles (Table 6): by 0–20% on ω_b, n_s and σ_8, the σ_8 ratio being above unity for both estimators. They disagreed on $\omega_{\rm cdm}$ (1.30 vs 0.94), and the absolute errors differed between the two estimators by a factor ~ 2 for $\omega_{\rm cdm}$, n_s, σ_8 and ~ 6 for ω_b (Table 7), as did the gain over the 2PCF: the regression is likely too optimistic (residual HOD contamination of its intercepts) and the emulator too conservative (training noise smoothing small slopes). The variants with auto-correlations or a single environment changed strongly between estimators and trainings.

Results with the independent covariance. With the same emulator derivatives, replacing the covariance by that of the 1786 independent boxes reversed the small lead of ASTRA split (Section 7). The 192 correlated samples misrepresented the relative noise of the two methods; the independent covariance is the more reliable one and is the baseline. What survived this change, and all the others, is that the two methods carry comparable information and are complementary.

Baseline. The baseline analysis was fixed after these steps: for the DESI-like mocks, the densified emulator trained on catalogue differences, the covariance of the official small boxes and the official HOD as fiducial point; for Quijote, the parameter subsets whose forecasts do not depend on the derivative noise. Other choices are treated as robustness tests.

7 Results

7.1 DESI-like mocks: ASTRA split, density split and the 2PCF

Table 8 and Figure 27 give the baseline forecasts. Both methods improve on the 2PCF by factors 1.6–3.1 depending on the parameter. Between them, the ratio $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ for the random-quartile cross-correlations is 0.90, 1.10, 0.84, 0.83, 0.94 and 0.92 for ω_b, $\omega_{\rm cdm}$, n_s, σ_8, w_0 and w_a: density split is slightly ahead, by 5–15%, except on $\omega_{\rm cdm}$. The two are clearly complementary: combined, they improve on the better one by 23% (σ_8) to a factor 2.4 (ω_b), 1.4 and 1.7 for w_0 and w_a, and the combination with the 2PCF is the most constraining (Figure 28). The errors on w_0 and w_a ($\simeq 0.8$ and 2.7

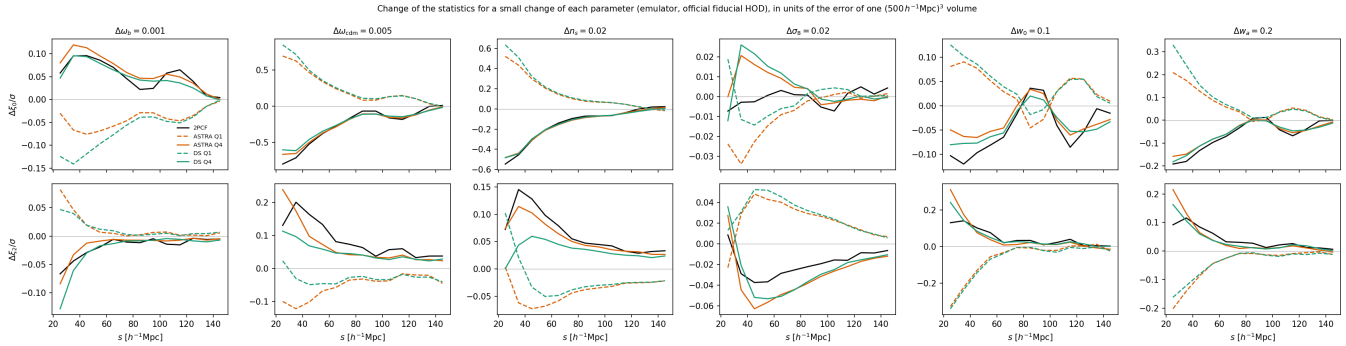


Figure 16: Change of the monopole (top) and quadrupole (bottom) for a small change of each parameter, in units of the error of one $(500\,h^{-1}\mathrm{Mpc})^3$ volume (baseline emulator at the official fiducial point): 2PCF (black), Q1 (dashed) and Q4 (solid) cross-correlations of ASTRA split (orange) and density split (green).

Change	Problem solved	Effect on ASTRA vs DS
Averaged HOD files → HOD-matched pairs	the 500 files are different HOD models: their scatter ($\sim 55\%$) swamped the cosmology ($\sim 1\%$)	derivatives were pure noise
Finite differences → emulator	the residual HOD mismatch of the pairs dominated the derivatives by a factor ~ 10; a calibrated HOD correction (8% accurate) was not enough	Fisher matrices became well defined
Emulator → densified emulator trained on catalogue differences	$\omega_{\rm cdm}$ response not learnt; derivatives of individual networks scattered by 70–170%; ranking changed between trainings	ranking stable between trainings; cross-checked by an independent regression over the derivative grid
192 subvolumes of one box → 1786 independent boxes	covariance from a single box, three correlated lines of sight	$\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ on σ_8: 1.12 → 0.85–1.11; ASTRA's apparent lead disappears
DESI-like: 4 → 6 cosmological parameters ($+w_0$, w_a)	the emulator and derivative grid allow the dark-energy parameters of the initial goals	errors on $\omega_{\rm cdm}$ (+40%) and n_s (+15%) degrade; ratios unchanged within errors
ASTRA classes → quartiles (main analysis), classes revisited with Quijote	with fixed thresholds, voids and knots contain $< 1\%$ of the points: two dominant classes	quartiles give balanced environments; classes + auto-correlations turn out the most constraining on Quijote
Quijote, 6 parameters → reliable subsets	derivative noise dominates the degenerate directions (Ω_b, h, M_ν)	ranking stable only for (Ω_m, σ_8), $(\Omega_m, n_s, \sigma_8)$

Table 5: The main changes of the analysis, the problem each solved, and its effect on the comparison.

quart.	deriv.	ω_b	$\omega_{\rm cdm}$	n_s	σ_8
random	emulator	1.02(15)	1.30(11)	1.11(9)	1.12(8)
random	regression	1.09(11)	0.94(4)	1.07(11)	1.16(8)
galaxy	emulator	0.80(9)	0.91(8)	0.92(7)	0.90(8)
galaxy	regression	1.08(16)	1.15(10)	1.10(13)	1.00(12)

Table 6: Superseded: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ with the 192-subvolume covariance (Q1+Q4 cross-correlations, HOD marginalised); above 1 favours ASTRA. In parentheses: uncertainty from the emulator ensemble or the bootstrap of the subvolumes.

data vector	ω_b	$\omega_{\rm cdm}$	n_s	σ_8
2PCF	.023/.0026	.018/.0067	.20/.077	.25/.083
ASTRA	.011/.0018	.015/.0071	.11/.051	.13/.081
Q1	.011/.0019	.019/.0067	.12/.054	.14/.094
Q4	.0075/.0014	.013/.0052	.090/.039	.11/.061
DS	.0066/.0011	.010/.0042	.081/.032	.10/.052
all				

Table 7: Superseded: marginalised errors with the 192-subvolume covariance, emulator / regression derivatives (random quartiles, compressed, HOD marginalised, one $(500\,h^{-1}\mathrm{Mpc})^3$ volume).

for one $(500\,h^{-1}\mathrm{Mpc})^3$ volume) are larger than the range of the AbacusSummit suite: at this volume the dark-energy parameters are not constrained, and only the relative information content of the statistics is meaningful. Adding them nevertheless keeps the ASTRA errors on the other parameters moderate (at most +40%, on $\omega_{\rm cdm}$) and leaves the DS/ASTRA ratios stable, so the comparison does not depend on whether w_0 and w_a are free. The variants (auto-correlations, galaxy quartiles, single environments) give ratios between 0.5 and 1.4 with no consistent winner (Figure 27).

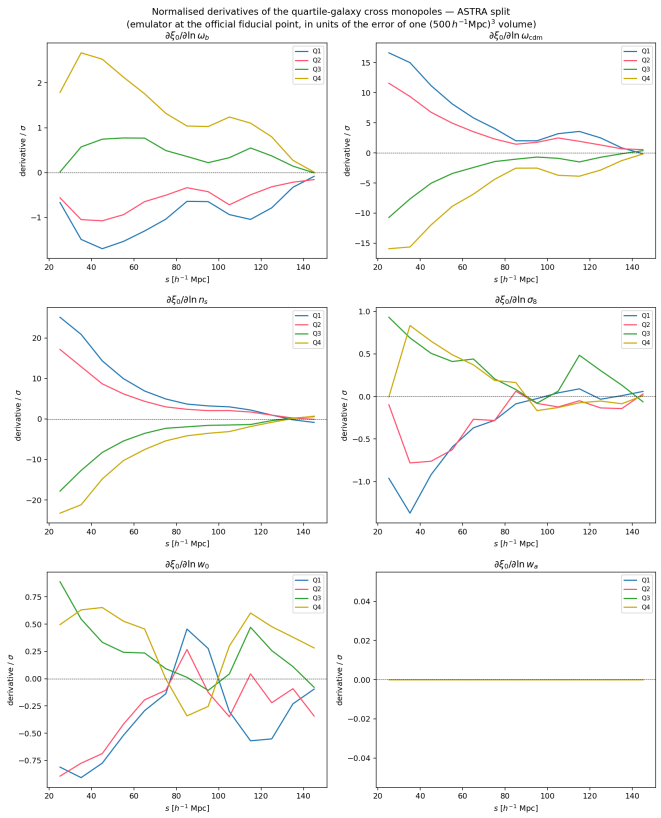


Figure 17: ASTRA split: derivatives of the four quartile–galaxy cross monopoles with respect to the logarithm of each parameter, in units of the error of one $(500\,h^{-1}\mathrm{Mpc})^3$ volume (baseline emulator, official fiducial point).

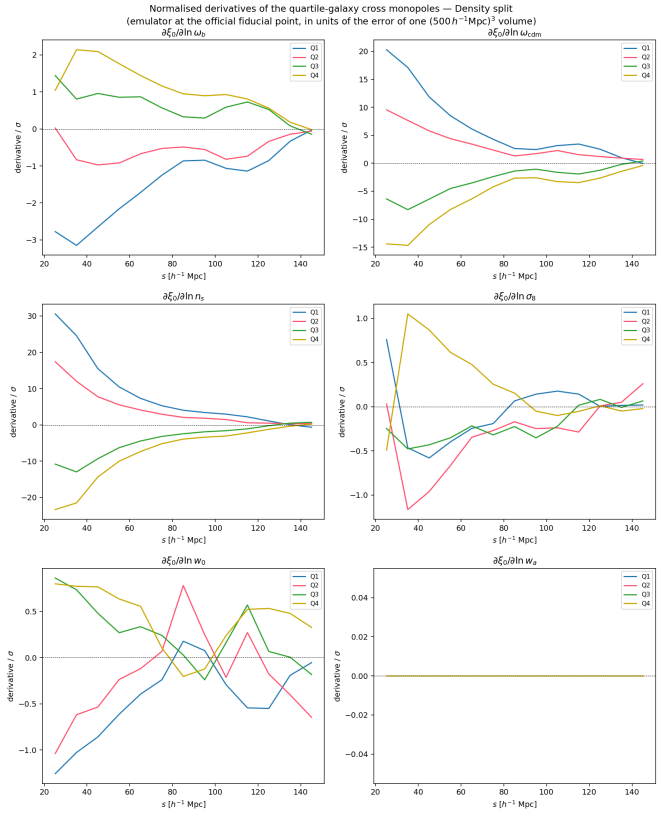


Figure 18: Same as Figure 17 for density split.

data vector	ω_b	$\omega_{\rm cdm}$	n_s	σ_8	w_0	w_a
2PCF	0.033	0.057	0.40	0.58	2.1	6.7
ASTRA	0.012	0.032	0.22	0.23	0.80	2.7
DS	0.011	0.035	0.18	0.19	0.75	2.5
ASTRA + DS	0.0046	0.022	0.13	0.15	0.53	1.5
2PCF + ASTRA + DS	0.0039	0.019	0.12	0.13	0.50	1.3

Table 8: Baseline marginalised 1σ errors (random-quartile Q1+Q4 cross-correlations, HOD marginalised) for one $(500\,h^{-1}\mathrm{Mpc})^3$ volume. These values are comparable to the prior widths and compare information content; they are not realistic constraints.

Figure 19 shows the corresponding contours: the environment statistics mainly reduce the ω_b–$\omega_{\rm cdm}$ and n_s–σ_8 degeneracies of the 2PCF.

Individual quartiles. Figures 20 and 21 show, as in the density-split analyses, the constraints of each quartile alone and of the four together, and Figures 22 and 23 compare the 2PCF, the void-like (Q1) and cluster-like (Q4) quartiles, all quartiles, and all quartiles with the 2PCF. A single quartile constrains the parameters about as well as the 2PCF, the intermediate quartiles (Q3 in particular) being the least constraining, but its information is of a different nature: the 2PCF averages the clustering over all environments, whereas a quartile measures it around one range of densities only, e.g. the profile of galaxies around voids for Q1. Each environment responds to the parameters in its own way: the void and cluster quartiles respond with opposite signs and different scale dependences (Figures 17, 18), and their degeneracy directions differ. The gain therefore comes from combining environments: the four quartiles together are several times tighter than any one of them or than the 2PCF, which mixes all environments and loses this distinction. Figure 24 summarises the information per environment: the Q4 cross-correlations alone already beat the 2PCF on most parameters, Q1 and Q4 together double the gain, and adding the auto-correlations multiplies it again.

Auto-correlations. Adding the quartile auto-correlations to the cross-correlations tightens the constraints of both methods by factors 1.3–2.8 depending on the parameter (Figure 25); with them, the two methods remain comparable (ratios 0.7–1.1), density split being ahead on ω_b, $\omega_{\rm cdm}$ and n_s and ASTRA on σ_8.

Scale cuts. Figure 26 shows the errors as a function of the minimum scale. All statistics lose information as small scales are removed, the environment

statistics staying 0.2–0.5 dex below the 2PCF up to $s_{\rm min} = 70\,h^{-1}$Mpc (beyond, the 2PCF has fewer bins than parameters). ASTRA and density split stay within 0.1 dex of each other at all scale cuts, their ranking changing with $s_{\rm min}$, and their combination keeps its 0.2–0.4 dex advantage.

Fiducial point. Evaluated at four different HODs, the ratio $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ stays between 0.77 and 1.14 for all six parameters (Table 9; 0.75–1.11 with four parameters): the fiducial point matters at the level of ±15%.

fiducial HOD	ω_b	$\omega_{\rm cdm}$	n_s	σ_8
official	0.90(11)	1.10(16)	0.84(15)	0.83(10)
`hod239`	0.83(9)	0.92(7)	1.04(12)	0.93(8)
`hod212`	0.77(7)	0.94(9)	0.94(13)	0.84(8)
`hod795`	0.79(8)	0.98(10)	1.06(16)	1.14(10)

Table 9: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ (above 1: ASTRA ahead) for the random-quartile cross-correlations with the official covariance at four fiducial HODs; in parentheses, the uncertainty from the emulator training.

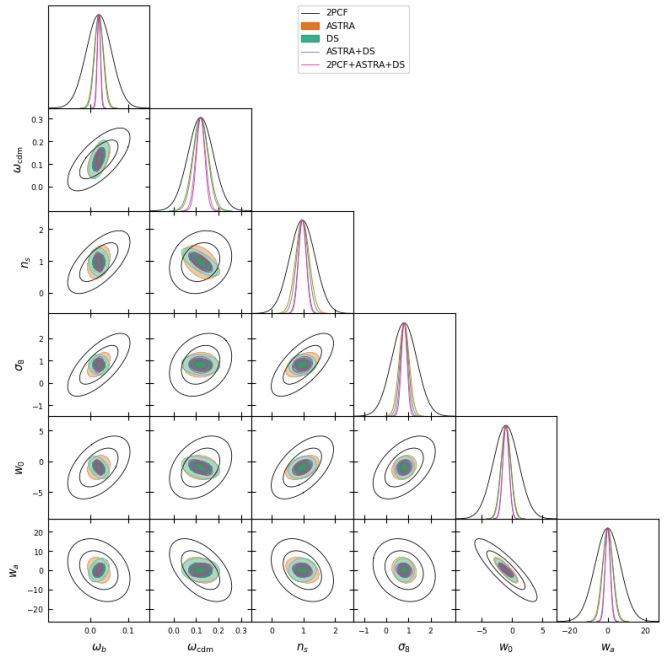


Figure 19: Baseline Fisher contours (68% and 95%) for the 2PCF, ASTRA split, density split and their combinations, one $(500\,h^{-1}\text{Mpc})^3$ volume, HOD marginalised.

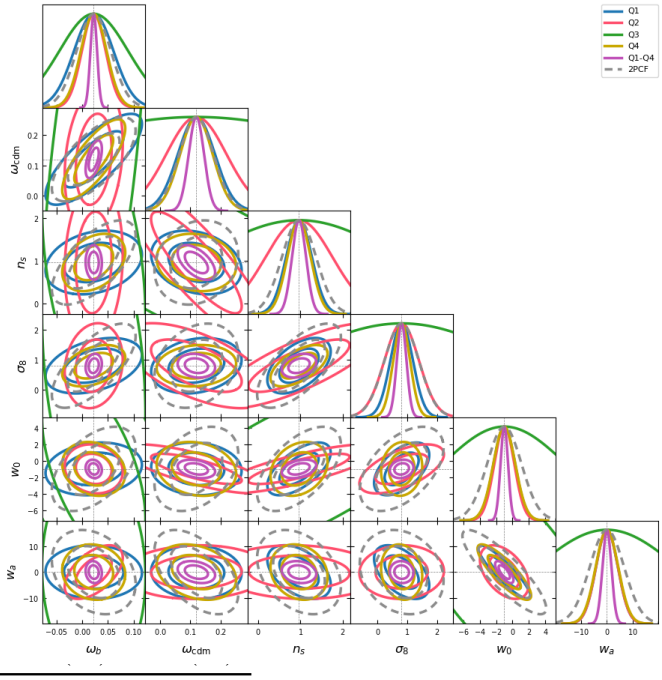


Figure 20: Constraints (68% and 95%) from individual and combined ASTRA-split quartiles (cross-correlations) and from the 2PCF (dashed), baseline, HOD marginalised. Axes span $\pm 3\sigma$ of the 2PCF.

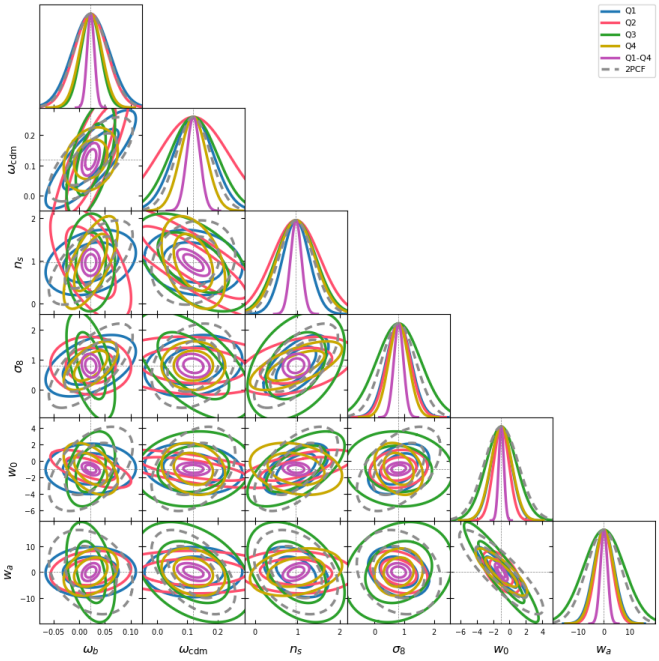


Figure 21: Same as Figure 20 for density split.

7.2 Quijote halos

Covariance. With 4000 fiducial realisations the covariance is converged: at fixed derivatives the marginalised errors change by less than 1% between 2000 and 4000 realisations, and by less than 6% between 1000 and 4000. (About half of the 15 517 fiducial catalogues of our copy were empty, an incomplete download without consequence given this convergence; all derivative sets are complete.)

Pairing. The ± realisations of a given index share their initial conditions: the scatter of the differences between realisations of the same index is 3–4 times smaller than between different indices, for the 2PCF

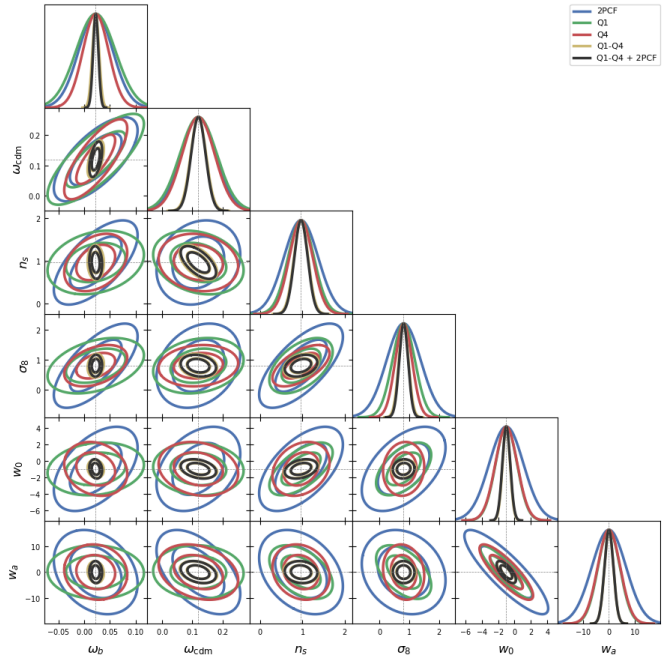


Figure 22: ASTRA split: constraints from the 2PCF, the Q1 and Q4 cross-correlations, all quartiles, and all quartiles with the 2PCF (baseline).

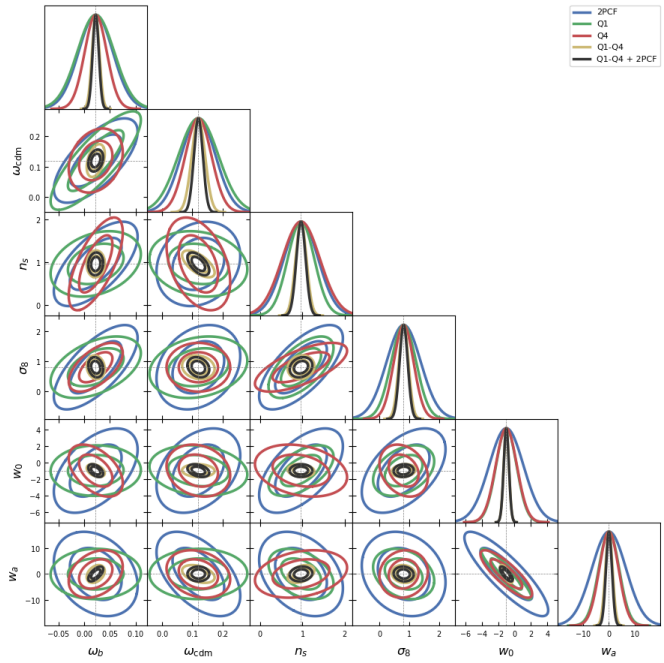


Figure 23: Same as Figure 22 for density split.

as for the quartile statistics.

Noise of the derivatives. Even paired, the derivatives remain noisy. Quijote is designed for statistics of the matter and halo fields (power spectrum, bispectrum, ...). Our observables add two sources of noise that the pairing cannot cancel:

1. *Sparse tracers.* With $\simeq 50\,000$ host halos per $(h^{-1}\mathrm{Gpc})^3$, each quartile contains only $\simeq 12\,000$ halos (or random points), and the shot noise of the quartile statistics is large.
2. *Discrete classification.* A small change of cosmology moves halos and random points from one quartile to another. This re-classification is only partly common to the + and − simulations, even with identical initial conditions, and adds a noise that does not cancel in the difference.

This noise is mostly amplified by the six-parameter Fisher matrix. Ω_b, h and n_s (and M_ν) are strongly degenerate in the clustering of halos: their weakest combination is constrained by a very small amount of information, of the same order as the noise left in the average of 500 derivative pairs, so that along this direction the derivatives are dominated by noise. The standard Fisher estimator is then biased high, and subtracting the expected noise term makes the six-parameter matrices non positive definite (Appendix B).

Estimator and reliability criterion. We therefore use the approach of [13]: the standard estimator is biased high by the derivative noise, a compressed estimator (half of the simulations define the compression, the other half measure its information) is biased low; the two bracket the true Fisher information, and their geometric mean is nearly unbiased. Where they agree, the forecast does not depend on the derivative noise: we call a forecast *reliable* when the compressed and standard errors differ by less than a factor 1.5 for both statistics compared.

Choice of the parameters. The bias grows with the length of the data vector and is concentrated in the degenerate directions. We computed the forecasts for six parameter subsets, from (Ω_m, σ_8) to all six parameters, and three compressions of the data vector ($10\,h^{-1}$Mpc bins with $\ell = 0, 2$; $20\,h^{-1}$Mpc bins with $\ell = 0, 2$; $20\,h^{-1}$Mpc bins, monopole only; Figure 29). Only (Ω_m, σ_8) and $(\Omega_m, n_s, \sigma_8)$ give reliable forecasts, for all compressions; as soon as Ω_b, h or M_ν are varied, the two estimators differ by factors 2–10 and the ranking of the methods changes from one compression to the next. These parameters cannot be constrained with these observables and 500 derivative pairs. Fixing Ω_b, h (and n_s), as many density-split forecasts do, is justified by external constraints (BBN, CMB); M_ν would require denser tracers or many more derivative simulations.

Results. The reliable forecasts (Table 10) agree with the DESI-like analysis for the random-quartile cross-correlations: density split is as constraining as ASTRA split or more ($\sigma_{\mathrm{DS}}/\sigma_{\mathrm{ASTRA}} = 0.6$–$1.1$). When the quartile auto-correlations are added, or

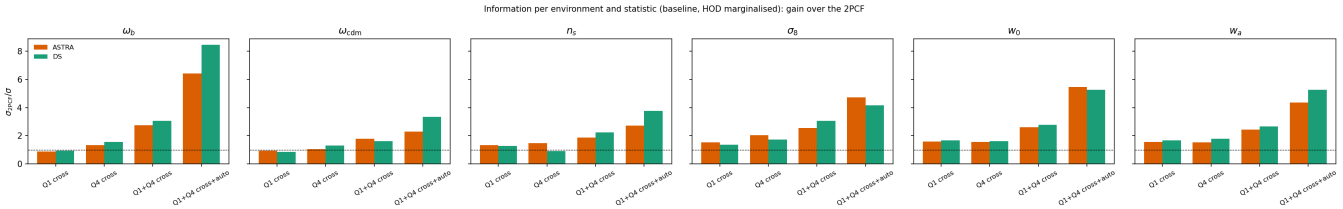


Figure 24: Gain over the 2PCF, $\sigma_{\rm 2PCF}/\sigma$, for the Q1 and Q4 cross-correlations alone, together, and with the quartile auto-correlations, for ASTRA split (orange) and density split (green) (baseline, HOD marginalised).

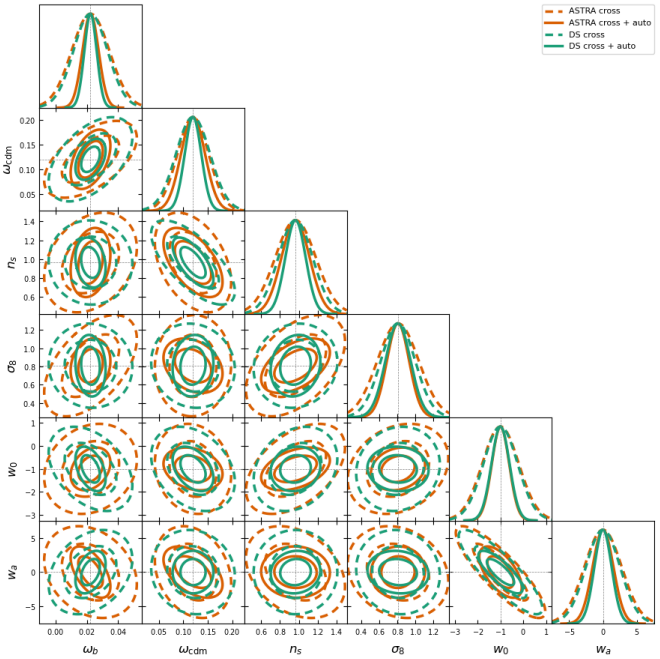


Figure 25: Q1+Q4 cross-correlations only (dashed) and with the quartile auto-correlations (solid), for ASTRA split (orange) and density split (green), baseline.

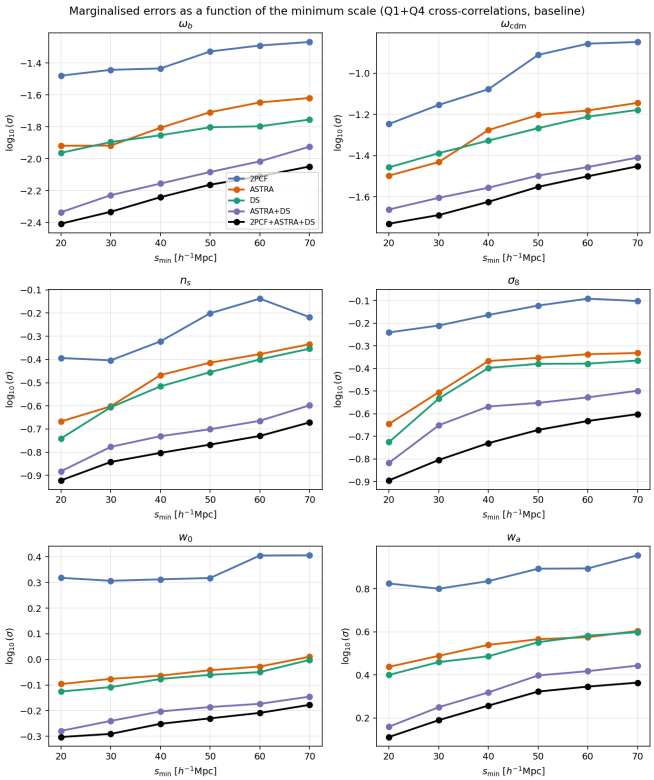


Figure 26: Marginalised errors as a function of the minimum scale $s_{\rm min}$ (Q1+Q4 cross-correlations, baseline).

with the halo quartiles, ASTRA split is more constraining on σ_8 by a factor 1.5–2 and comparable on Ω_m and n_s: its information lies more in the clustering of its environments with themselves and in the environments of the tracers than in the tracer profiles around random environments.

quartiles	parameters	Ω_m	n_s	σ_8
random, cross	Ω_m, σ_8	0.86–1.04	–	0.65–0.81
random, cross	Ω_m, n_s, σ_8	0.57–0.64	0.68–0.74	1.04–1.14
random, + auto	Ω_m, σ_8	1.19–1.40	–	1.54–2.11
random, + auto	Ω_m, n_s, σ_8	0.74–0.81	0.85–0.92	1.72–1.78
halo, cross	Ω_m, σ_8	1.18–1.21	–	1.91–2.04
halo, cross	Ω_m, n_s, σ_8	1.16–1.20	1.17–1.23	1.84–1.90

Table 10: Quijote, reliable forecasts: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$, range over three compressions of the data vector (above 1: ASTRA ahead).

7.3 ASTRA cosmic-web classes (Quijote)

To test whether the original ASTRA classes carry more information than quartiles, we measured them on Quijote, where the derivatives are clean: 4000 fiducial realisations for the covariance and the 500 pairs of Ω_m, n_s and σ_8, with the same halos, random points and estimators as the quartile analysis, and compared them with the quartiles of ASTRA and density split measured on the same realisations ($10\,h^{-1}$Mpc bins, $\ell = 0, 2$, $s \geq 20\,h^{-1}$Mpc). Because the void and knot classes are nearly empty for random points, the class data vectors use the sheets and filaments (adding the voids changes nothing).

Class fractions. The fractions of points in each class are themselves sensitive to cosmology (Figure 30): a variation of Ω_m, n_s or σ_8 shifts the sheet

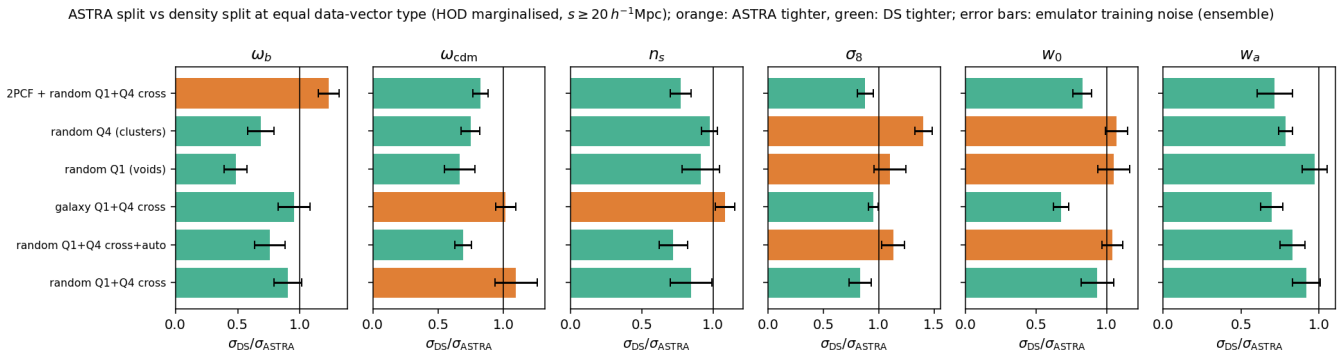


Figure 27: Baseline: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ for several data-vector types (above 1: ASTRA split more constraining). Error bars: uncertainty from the emulator training.

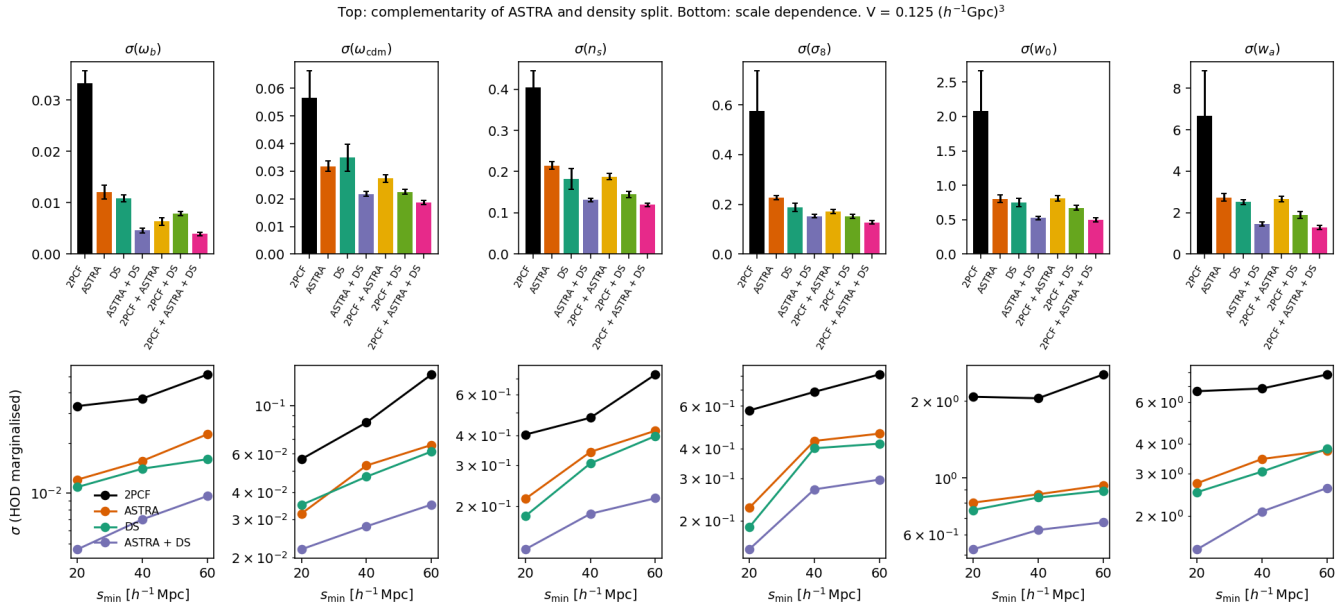


Figure 28: Baseline. Top: marginalised errors of the 2PCF, ASTRA split, density split and their combinations. Bottom: dependence on the minimum scale.

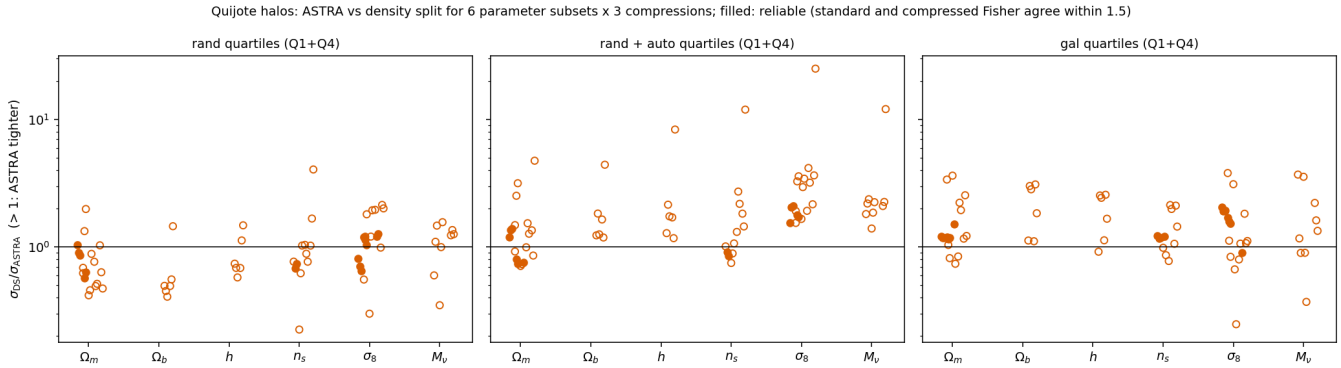


Figure 29: Quijote halos: $\sigma_{\rm DS}/\sigma_{\rm ASTRA}$ (above 1: ASTRA more constraining) for six parameter subsets and three compressions, for the random-quartile cross-correlations (left), with their auto-correlations (centre), and the halo quartiles (right). Filled: reliable forecasts (standard and compressed Fisher estimators within a factor 1.5); open: dominated by the noise of the derivatives.

and filament fractions of the random points by 0.3–1 times their scatter between realisations, in opposite directions. Unlike quartiles, which have fixed fractions by construction, fixed thresholds let the volume of each environment respond to cosmology.

Information. Table 11 and Figure 31 compare the classes with the quartiles of ASTRA and density split for the three compressions of the data vector used in Section 7.2. For (Ω_m, σ_8) all the forecasts are reliable. With cross-correlations only, the classes carry about as much information as the ASTRA quartiles, and less than density split on σ_8. With the auto-correlations, the classes are the most constraining statistic for every compression: $\sigma(\sigma_8)$ is 2.1–3.0 times smaller than with density split and 1.35–1.45 times smaller than with the ASTRA quartiles, and $\sigma(\Omega_m)$ 1.3–1.5 and ~ 1.1 times smaller. For $(\Omega_m, n_s, \sigma_8)$, the classes with auto-correlations are reliable for all compressions (and are the only reliable ASTRA random-point vector with the finest binning); their advantage over density split remains a factor 2.8–2.9 on σ_8, but shrinks with the compression on Ω_m (1.0–1.5) and n_s (1.2–1.85). The classes of the halos give $\sigma(\Omega_m)$ 15% smaller than the halo quartiles but a larger $\sigma(\sigma_8)$.

compression	Ω_m	n_s	σ_8
(Ω_m, σ_8)			
$10\,h^{-1}$Mpc, $\ell = 0, 2$	1.28	–	2.08
$20\,h^{-1}$Mpc, $\ell = 0, 2$	1.49	–	2.95
$20\,h^{-1}$Mpc, $\ell = 0$	1.52	–	2.96
$(\Omega_m, n_s, \sigma_8)$			
$10\,h^{-1}$Mpc, $\ell = 0, 2$	1.51*	1.85*	2.88*
$20\,h^{-1}$Mpc, $\ell = 0, 2$	1.07	1.31	2.86
$20\,h^{-1}$Mpc, $\ell = 0$	0.99	1.20	2.78

Table 11: Quijote, 4000 fiducial realisations: $\sigma_{\rm DS}/\sigma_{\rm classes}$ for the random-point cross- and auto-correlations of the density-split quartiles and of the ASTRA classes (sheets and filaments); above 1: classes more constraining. *: density split marginally reliable (estimator ratio 1.5–1.6).

Interpretation. The gain comes from the auto-correlations of the two classes and from the response of their volumes to cosmology, which equal-size quartiles remove by construction. The advantage on σ_8, a factor 2–3 over density split, is robust to the compression, the parameter set and the number of covariance realisations; the advantage on Ω_m and n_s is smaller and depends on the compression. For a DESI-like analysis, this would require an emulator trained on class statistics, i.e. remeasuring the emulator suite.

7.4 Inference with the emulator (MCMC)

Fisher forecasts assume Gaussian posteriors and are local. To test the full pipeline as it would be applied to data, we sampled the posterior of the 20 parameters (8 cosmological, 12 HOD) with the emulator as the model: Gaussian likelihood with the official covariance rescaled to the data volume plus the emulator error, uniform priors over the training range, `emcee` with 64 walkers and 8000 steps (Appendix B). The data are measurements the emulator has never seen: one subvolume of each of 2 catalogues of the 8 test cosmologies, and 8 subvolumes of the official base box of the fiducial cosmology, each analysed with the 2PCF, ASTRA split, density split and their combinations (120 chains).

The posteriors are unbiased and calibrated or slightly conservative: the truth lies within 1σ of the posterior mean in 70–100% of the runs depending on the parameter (68% expected). They are however barely narrower than the priors (Figures 32 and 33): one $(500\,h^{-1}\text{Mpc})^3$ volume with 12 free HOD parameters constrains only $\omega_{\rm cdm}$ and σ_8 beyond the prior, by 10–15%, and the statistics differ by a few per cent. A first set of runs on the full $(2\,h^{-1}\text{Gpc})^3$ box was limited the other way: the emulator error (0.4–0.9 times the statistical error of that volume) was as large as the statistical error, and masked the differences between the statistics. Distinguishing the statistics through inference therefore requires both a large volume and an emulator error well below the statistical error of that volume; with the present training set this is not the case (and the learning curve shows that more catalogues of the same kind would not change it), so the comparison of the methods rests on the Fisher forecasts. The inference pipeline is validated and ready for a larger or less noisy training set.

8 Robustness tests

We test the sensitivity of the measurements to the numerical choices on the first 8 subvolumes of the fiducial catalogue (line of sight z). Each test is repeated for both methods, and the change of the Q1 and Q4 cross-correlation monopoles is expressed in units of the statistical error of one subvolume (Figure 34).

Periodic padding. Increasing the padding around each subvolume from 30 to $60\,h^{-1}$Mpc leaves the ASTRA values of r, and therefore all the statistics, unchanged: the Delaunay neighbourhood of every point of the subvolume is already contained in the $30\,h^{-1}$Mpc padding.

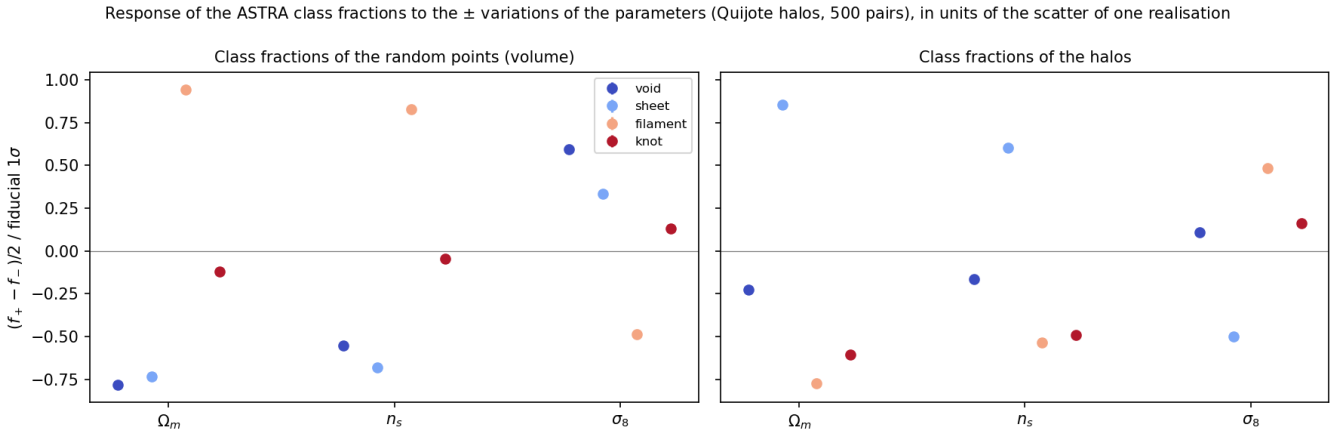


Figure 30: Response of the ASTRA class fractions of the random points (left) and halos (right) to the ± variations of Ω_m, n_s and σ_8 (Quijote, 500 pairs), in units of the scatter of one realisation (error bars: error of the mean, smaller than the symbols).

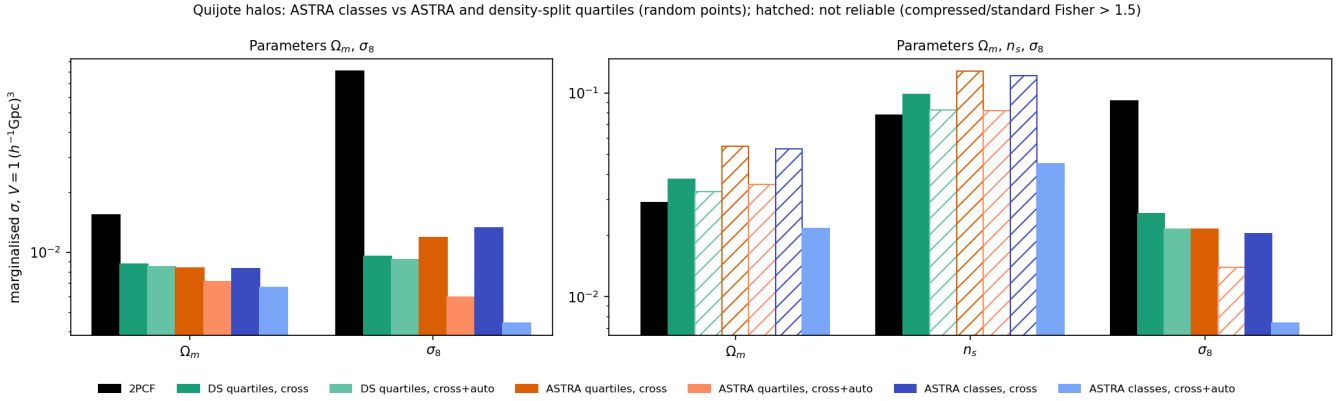


Figure 31: Quijote halos, $V = 1\,(h^{-1}\mathrm{Gpc})^3$: marginalised errors for (Ω_m, σ_8) (left) and $(\Omega_m, n_s, \sigma_8)$ (right) from the 2PCF, the density-split and ASTRA quartiles of the random points, and the ASTRA classes (sheets and filaments), with cross-correlations only or with auto-correlations. Hatched: not reliable (compressed and standard Fisher estimators differ by more than a factor 1.5).

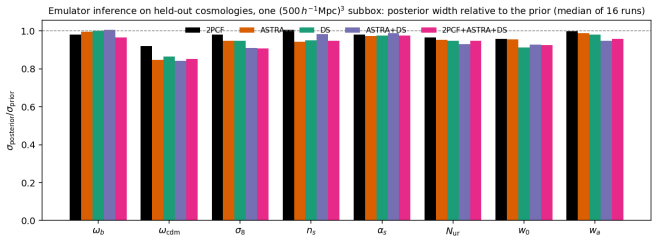


Figure 32: Emulator inference on held-out cosmologies (one $(500\,h^{-1}\mathrm{Mpc})^3$ volume, official covariance): posterior width relative to the prior width, median of 16 runs, for the 2PCF, ASTRA split, density split and their combinations.

Number of randoms. Doubling the number of random points in the tessellation ($N_r = 2N_g$) shifts the ASTRA Q1 cross-correlations by up to 4σ and the Q4 ones by up to 2σ at $s \lesssim 40\,h^{-1}$Mpc, while density split, for which the randoms are only query points, changes by less than 0.2σ. In ASTRA the randoms enter the definition of r itself (Eq. 1), so N_r/N_g is a parameter of the statistic and must be kept fixed.

Sampling density. Randomly thinning the galaxies (and the randoms in the same proportion) by a factor 2 (5) shifts the ASTRA cross-correlations by 7–10 (20–35)σ at $s \simeq 20$–$40\,h^{-1}$Mpc, against ~ 1 (3–5)σ for density split. At the level of individual points, the rank correlation between the r of the same galaxies computed in the full and in the thinned samples is only 0.65, 0.38 and 0.24 for thinning factors of 2, 5 and 10, and only 58%, 42% and 36% of the galaxies of the extreme quartiles keep their quartile.

This is an intrinsic property of the estimator rather than a numerical instability. ASTRA is an *adaptive-scale* statistic: the environment of a point is defined by its Delaunay neighbours, so its effective scale is the typical distance between neighbours, $\propto \bar{n}^{-1/3}$, and it changes with the sampling density. Thinning enlarges this scale, which is why the changes are concentrated at $s \lesssim 50\,h^{-1}$Mpc, whereas density split

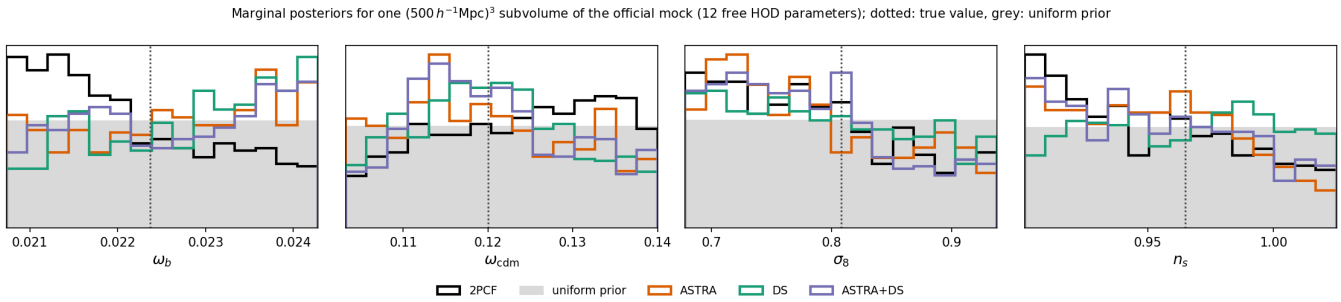


Figure 33: Marginal posteriors of four cosmological parameters for one $(500\,h^{-1}\mathrm{Mpc})^3$ subvolume of the official mock (fiducial cosmology, dotted lines), with 12 free HOD parameters, for the 2PCF, ASTRA split, density split and ASTRA + DS. At this volume all posteriors remain close to the uniform prior (grey).

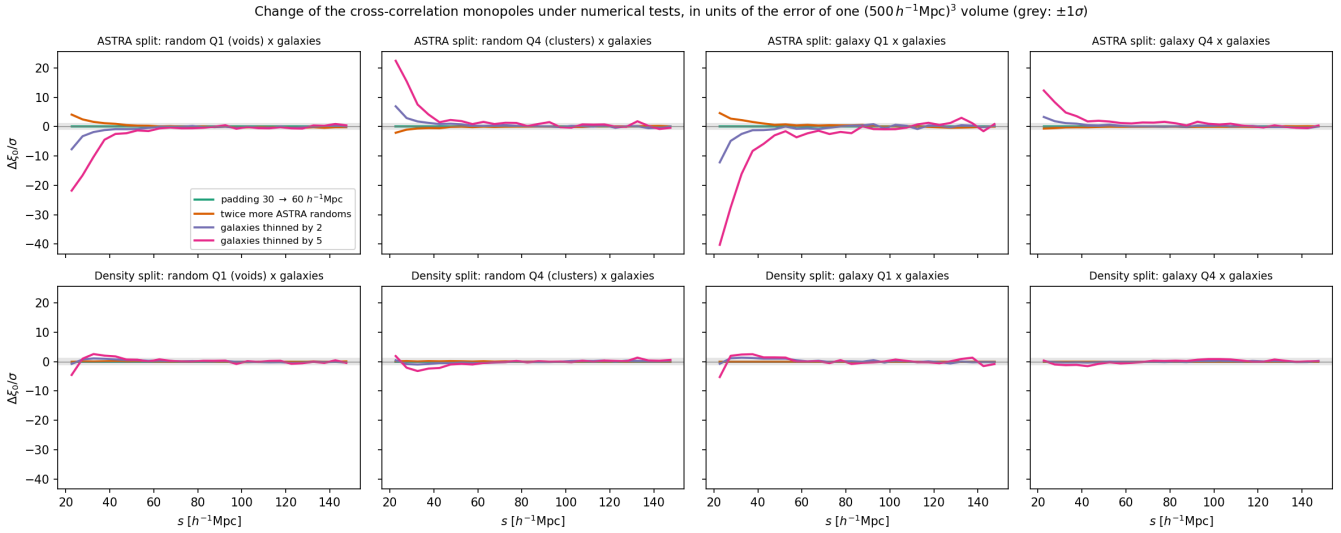


Figure 34: Change of the Q1 and Q4 cross-correlation monopoles (random quartiles, left; galaxy quartiles, right) of ASTRA split (top) and density split (bottom) when the padding is doubled (green), the ASTRA randoms are doubled (orange), and the galaxies are thinned by 2 (purple) and 5 (pink), relative to the baseline, in units of the error of one $(500\,h^{-1}\mathrm{Mpc})^3$ volume (grey band: $\pm1\sigma$). Mean over 8 subvolumes of the fiducial catalogue.

keeps a fixed smoothing radius. This is not a limitation for the present analysis or for DESI, since the density is fixed in the mocks as in the data (and in an emulator trained at that density). It implies however that ASTRA statistics must be compared, modelled and emulated at the same number density and random-to-galaxy ratio, and that comparisons with density split must be made at equal density, as throughout this paper.

9 Discussion

What is robust. The comparison was made with three derivative estimators (emulator, regression over pairs, Quijote finite differences), two covariances, four fiducial HODs, two tracer populations (LRG-like galaxies, halos) and several data vectors. Two conclusions never changed: ASTRA split and density split carry comparable information (error ratio 0.75–1.2 for the same data vector, with four or six parameters, in the DESI-like analysis with the independent covariance, 0.6–2 in the reliable Quijote forecasts), and they are complementary, their combination gaining 23% to a factor 2.4. The ranking for individual parameters changed with the analysis choices (Table 5), at the 10–40% level, which is the size of the effect of those choices; we therefore do not claim a winner. Where a trend is consistent, density split is slightly ahead on ω_b, Ω_m and n_s with the random-quartile cross-correlations, and ASTRA split ahead on σ_8 through its environment auto-correlations and tracer environments (Quijote). In particular, the advantage of ASTRA split on σ_8 found with the single-box covariance disappears with the independent covariance.

What we learnt about the methodology. (i) Emulator suites in which the HOD is drawn independently in each cosmology do not allow naive finite differences: the HOD mismatch dominates the

cosmological response by an order of magnitude, and paired differences do not cancel the galaxy sampling noise of different HOD catalogues. (ii) Single-box covariances, even with many subvolumes and lines of sight, can bias the *comparison* of statistics, not only their absolute errors. (iii) Environment statistics of sparse tracers have noisy derivatives, which a six-parameter Fisher matrix amplifies along its degenerate directions; comparing two Fisher estimators that bracket the truth identifies the reliable forecasts. (iv) The clustering statistics respond essentially to a single combination of the 12 HOD parameters.

Limitations. (i) The forecasts are for one $(500\,h^{-1}\mathrm{Mpc})^3$ volume, where the errors are comparable to the prior widths: they compare information content and are not realistic constraints. (ii) The inference is calibrated but not yet discriminating: at this volume the prior dominates, and for a full box the emulator error does. (iii) The Quijote halos are ten times sparser than the LRGs, and ASTRA's adaptive scale is correspondingly larger. (iv) The emulator suite has a single HOD realisation per point, whose galaxy sampling noise sets the floor of the emulator error.

Improvements. A learning curve shows that the emulator error has reached a plateau at about half the statistical error of a $2\,h^{-1}$Gpc box: more training catalogues of the same kind would barely help. Several HOD realisations (seeds) per point, or HOD-matched catalogues generated with AbacusHOD in all cosmologies, are needed both to lower this floor, which would make an inference on a DESI-like volume discriminating, and to obtain clean derivatives in the DESI-like setting. Larger sets of independent boxes would remove the remaining dependence on the covariance.

Beyond quartiles. Being local and free of smoothing, ASTRA provides more than a density ranking: a classification of every galaxy into voids, sheets, filaments and knots, a continuous environment value usable as a mark, and a natural treatment of survey masks through the random points. The first test of the classes on Quijote (Section 7.3) is the most promising result of this work for ASTRA: despite their imbalance, the classes with their auto-correlations constrain σ_8 two to three times better than density split, robustly across compressions. Natural next steps are the probabilistic classes of [4] (classification repeated over many random catalogues, which gives every point a probability of belonging to each class and could make the void and knot classes usable), thresholds adapted to the density of the sample, and an emulator of class statistics for DESI.

10 Conclusions

We set out to measure how much cosmological information ASTRA environment statistics add to the galaxy two-point function, and how they compare with density split. We measured both on the same DESI-like mocks and Quijote halos, built covariance and derivative datasets for Fisher forecasts, and tested the full chain with an emulator-based inference. The main points are:

- **Gain over the two-point function.** ASTRA environment statistics constrain ω_b, ω_{cdm}, n_s, σ_8, w_0 and w_a 1.6–3.1 times better than the galaxy two-point function.
- **Comparison with density split.** The two methods carry comparable information (error ratio 0.8–1.1, density split slightly ahead) and are complementary: combining them gains another 23% to a factor 2.4. With Quijote halos, ASTRA does better on σ_8 (factor 1.5–2) through its environment auto-correlations and tracer environments.
- **Cosmic-web classes.** With the fixed ASTRA thresholds, the classes reduce in practice to two dominant classes (sheets and filaments) at these densities, which is why the main analysis uses quartiles. On Quijote, the classes with their auto-correlations are nevertheless the most constraining statistic: two to three times better than density split on σ_8, robustly across compressions.
- **What ASTRA measures.** The ASTRA and density-split environments are strongly but not perfectly correlated ($\rho \simeq 0.72$). The ASTRA value r itself hardly responds to cosmology: the information lies in how the environments cluster. ASTRA adapts its scale to the local density, so it must be modelled at the density and random-to-galaxy ratio of the data.
- **Method.** Emulator suites with an independent HOD per cosmology do not allow naive finite differences, single-box covariances can bias the comparison of statistics, and environment statistics of sparse tracers have noisy derivatives. We showed how each problem was diagnosed and addressed. The emulator inference is unbiased and calibrated, but discriminating between statistics with it would require several HOD realisations per training point.

Natural continuations, beyond the scope of this paper, are the probabilistic ASTRA classes of [4], which could make the void and knot classes usable, and an emulator of the class statistics for a DESI-like analysis of the classes.

A Reproducing this paper

Code. All the measurements, forecasts and figures are produced by the code at `https://github.com/Hugoooat/DESI_astra/tree/astra_hugo_v2`, run on NERSC Perlmutter CPU nodes with the DESI `cosmodesi` environment:

```
source /global/common/software/desi/users/
    adematti/cosmodesi_environment.sh main
```

The code is organised in three modules and a set of scripts: `astra_ds.py` contains the whole measurement of one catalogue (reading, periodic padding, ASTRA tessellation, density field, quartiles, subvolumes, clustering), `analysis.py` the loading of measurements, data vectors, covariances and Fisher tools, `emulator.py` the emulator (networks, training with the difference loss, derivatives). Each catalogue is measured on one 128-core node in $\simeq 8$ minutes (64 subvolumes processed in parallel, ASTRA and density split, 33 statistics, three multipoles); measurements are saved as one `.npz` file per catalogue, line of sight and space, containing the multipoles of every statistic for every subvolume. Lists of catalogues are run with `slurm/run_list.sh` inside an allocation (one `srun` per node).

Data. The three data sets of Table 1 are read from the following locations on NERSC:

- **AbacusSummit emulator suite**: `/pscratch/sd/n/ntbfin/emulator/hods/z0.5/` (85 cosmology directories, `hod000`–`hod499`); cosmological parameters in `config/cosmologies.csv`. The catalogues measured are listed, one line per catalogue (cosmology, HOD, space, line of sight), in `config/runs_emulator*.txt` (8 HODs per cosmology), `runs_hod_calibration.txt` (72), `runs_local_*.txt` (240), `runs_extra.txt` (54) and `runs_real_space.txt` (27); the fiducial catalogue and the HOD-matched pairs are in `config/hod_selection.json`.
- **Official DESI mocks**: `/global/cfs/cdirs/desi/cosmosim/AbacusHOD_mocks/v1/CubicBox/LRG/z0.800/catalog_fba_kp2d_lrg_main` (1786 files) and `.../CubicBox/LRG/z0.500/AbacusSummit_base_c000_ph{000--024}/`.
- **Quijote**: Rockstar halo catalogues `{set}/{i}/out_3_pid.list` for the sets `fiducial` (realisations 0–3999), `fiducial_ZA`, `Om_p/m`, `Ob2_p/m`, `h_p/m`, `ns_p/m`, `s8_p/m`, `Mnu_p/pp/ppp` (realisations 0–499), transferred with Globus from the Quijote public data to `/pscratch/sd/h/hugon/quijote/`, following the instructions of the Quijote documentation (`https://quijote-simulations.readthedocs.io`).

Intermediate products (measurements, trained emulators, chains; $\simeq$ 3 GB) are stored in `/pscratch/sd/h/hugon/astra-ds-store/` and are regenerated by the steps below.

Steps.

```
python scripts/select_hods.py; python scripts/
    select_runs.py; python scripts/
    select_local_runs.py
salloc ... bash slurm/run_list.sh config/runs_
    *.txt # measurements
sbatch slurm/night_stage1.sbatch # official
    mocks + Quijote
python scripts/build_emulator_dataset.py
python scripts/build_pair_dataset.py --holdout
    0 --out pair_dataset_full.npz
python scripts/train_emulator.py --nseeds 10 --
    pairs data/pair_dataset_full.npz --models-
    dir models_pairs --tag _pairs
python scripts/compare_methods.py --models-dir
    models_pairs --validation-tag _pairs --
    compress --cov covbox --fiducial official
    --params omega_b omega_cdm n_s sigma8 w0 wa
python scripts/quijote_fisher.py --params Om s8
     --rebin 2 --tag _p0c0; python scripts/
    summarize_quijote.py
python scripts/mcmc.py --data official:000:0 --
    vector ASTRA --cov covbox --volume-factor
    1; python scripts/summarize_mcmc.py
bash slurm/quijote_classes.sh # ASTRA classes
    on Quijote (run_quijote.py --classes)
python scripts/quijote_classes_fisher.py --
    params Om s8 --tag _p2; python scripts/
    classes_figures.py
python scripts/robustness.py; python scripts/
    plot_real_space.py
python scripts/paper_figures.py --params
    omega_b omega_cdm n_s sigma8 w0 wa
python scripts/paper_figures_v1style.py --
    params omega_b omega_cdm n_s sigma8 w0 wa
python scripts/readable_figures.py; python
    scripts/env_maps.py; python scripts/
    classes_figures.py
cd paper && make # article, article_fr,
    vulgarisation, poster
```

The full set of measurements costs $\simeq 200$ node-hours (1043 emulator-suite catalogues, 1786 small boxes and 25 base boxes, 11 000 Quijote realisations at $\simeq 5$ s each).

Setting	Value
Subvolumes	4^3 of $500\,h^{-1}$Mpc
ASTRA padding, randoms	$30\,h^{-1}$Mpc periodic, $N_r = N_g$
DS smoothing, mesh	Gaussian $10\,h^{-1}$Mpc, $5\,h^{-1}$Mpc CIC
Quartiles	4, equal counts
Geometry randoms	$5N_g$ per subvolume
s, μ bins	$5\,h^{-1}$Mpc in $[0, 150]$, 120 in $[-1, 1]$
Multipoles	$\ell = 0, 2, 4$
Data vectors	$\ell = 0, 2$, $s \in [20, 150]$, $10\,h^{-1}$Mpc
Seed	42

Table 12: Numerical settings.

B Mathematical details

Fisher matrix and Hartlap factor. For a data vector $\boldsymbol{d}$ with covariance C estimated from N_s samples, $F_{ij} = \partial_i \boldsymbol{d}^T \hat{C}^{-1} \partial_j \boldsymbol{d}$ with $\hat{C}^{-1} = \frac{N_s - N_d - 2}{N_s - 1} C^{-1}$ [14]; the marginalised errors are $\sigma_i = \sqrt{(F^{-1})_{ii}}$ and the errors at fixed other parameters $1/\sqrt{F_{ii}}$. HOD parameters are marginalised by adding a Gaussian prior P_h to the HOD block and inverting the full matrix.

Compression. $\boldsymbol{t} = W^T \boldsymbol{d}$, $W = \tilde{C}^{-1} D$, with D the derivatives with respect to the 16 parameters and $\tilde{C}$ a Ledoit–Wolf shrinkage covariance. If $\tilde{C} = C$, $\boldsymbol{t}$ keeps all the Fisher information of $\boldsymbol{d}$; otherwise it keeps less, without bias. The covariance of $\boldsymbol{t}$ is estimated from the samples, with a Hartlap factor close to unity.

Noise of the derivatives. Derivatives estimated as an average over N pairs are $\hat{D} = D + \epsilon$, with a noise of covariance Σ_D / N. The standard Fisher estimator then has the expectation

$$\langle \hat{D}^T C^{-1} \hat{D} \rangle = D^T C^{-1} D + \mathrm{Tr}\big(C^{-1} \Sigma_D\big)/N. \qquad (3)$$

The second term is a positive bias: the noise adds spurious information. It is roughly the same in all directions of parameter space, but matters only where the true information $D^T C^{-1} D$ is small, i.e. in the degenerate directions; the marginalised errors, set by the inverse of the matrix, are controlled precisely by these weak directions. We use three tests:

1. the noise fraction $\mathrm{Tr}(C^{-1}\Sigma_D)/(N F_{ii})$; above 20–30% the forecast is suspect;

2. the bracketing of [13]: the standard estimator is biased high; a compressed estimator, in which half of the simulations define the compression and the other half measure its information, is biased low. If both give the same error the noise does not matter; if they differ by factors 2–10 the result reflects the noise, not the physics. We call a forecast reliable when they agree within a factor 1.5, and quote their geometric mean, which is nearly unbiased;

3. convergence: the forecast is recomputed with $N/2$ pairs.

Volume. The forecasts are computed for one $(500\,h^{-1}\mathrm{Mpc})^3$ volume, where the absolute errors are comparable to the prior widths; this is the nature of a comparison of information content and is not corrected. The ratios $\sigma_{\mathrm{DS}}/\sigma_{\mathrm{ASTRA}}$ do not depend on the volume as long as the covariance scales as $1/V$ and the emulator error is negligible, which holds at this volume.

Emulator loss. With outputs standardised by the statistical error of a $2\,h^{-1}$Gpc box σ_{box},

$$\mathcal{L} = \Big\langle \frac{(f(x_i) - y_i)^2}{\sigma_{\mathrm{box}}^2} \Big\rangle + \lambda \Big\langle \frac{[(f(x_i) - f(x_j)) - (y_i - y_j)]^2}{\sigma_{ij}^2 + \epsilon^2} \Big\rangle, \qquad (4)$$

where the second average runs over the 2544 pairs of catalogues, σ_{ij} is the error of their difference (64 paired subvolumes), $\lambda = 1$, $\epsilon = 0.05\,\sigma_{\mathrm{box}}$.

Regression over pairs. $\boldsymbol{d}_c(h) = \boldsymbol{a}_c + G\,V(h - h_0)$ for $c = c_\pm$, with a shared response G and V the 3 leading directions of the HOD space (singular vectors of the HOD response of the 72 fiducial catalogues); derivative $(\boldsymbol{a}_+ - \boldsymbol{a}_-)/(\theta_+ - \theta_-)$; errors by bootstrap over subvolumes.

Inference. Gaussian likelihood $-2 \ln \mathcal{L} = (\boldsymbol{d} - \boldsymbol{f}(\boldsymbol{\theta}, \boldsymbol{h}))^T (C/V_{\mathrm{f}} + C_{\mathrm{emu}})^{-1} (\boldsymbol{d} - \boldsymbol{f})$, with C the official covariance, V_{f} the data volume in units of $(500\,h^{-1}\mathrm{Mpc})^3$ and C_{emu} the emulator error; uniform priors over the training range; `emcee` with 64 walkers and 8000 steps. Calibration is tested by the fraction of runs whose truth lies within 1σ of the posterior mean (68% expected).

C Methodological path and intermediate results

This appendix details the steps of Table 5 in chronological order.

1. HOD files are not realisations. A first version of the pipeline averaged the statistics of 64 different HOD files per cosmology to build "covariance" and "derivative" sets, and measured them in the central $500\,h^{-1}$Mpc cube of each box. The HOD files are different HOD models, not realisations: their scatter ($\sim 55\%$) dominated the cosmological differences

($\sim 1\%$), and the finite differences were compatible with noise (median $|S/N| < 1$ per bin for ω_b, n_s, σ_8). That version also had five implementation problems, all corrected: pair counts assumed periodic boundary conditions in a non-periodic cut box; the 2PCF used the mid-point line of sight from the centre of the box instead of the axis along which the RSD were applied, which erased the quadrupole; quartiles were computed jointly on galaxies and randoms, giving unequal and realisation-dependent sizes; the Delaunay triangulation was computed on the cut box without periodic padding; and ASTRA was first applied to the random catalogue alone, so that the random quartiles carried no information on the galaxy field and their correlations were consistent with zero.

2. HOD-matched pairs. We then selected, for each $\pm$ pair, the two catalogues with the closest HODs (twice closer than random pairs, Table 3) and differenced them subvolume by subvolume (Section 5.2). The differences are precise but dominated by the residual HOD mismatch: a 2% change of ω_b implied tens of per cent changes of the 2PCF at $s \sim 20\,h^{-1}$Mpc, and the Fisher matrices were singular.

3. HOD calibration. The HOD response measured on 72 fiducial catalogues and subtracted from the pairs was predicted with a 45% (linear) and 8% (GP) error; with an HOD term ten times the signal, even 8% leaves a residual of the size of the signal.

4. Emulator. The first emulator, trained on 8 HOD draws per cosmology, reproduced held-out cosmologies to a fraction of the statistical error, but did not capture the response to $\omega_{\rm cdm}$ below $70\,h^{-1}$Mpc (Figure 13). A preliminary comparison with this emulator gave ASTRA and density split comparable overall, density split ahead in the voids (Q1) for ω_b and $\omega_{\rm cdm}$, and their combination 15–25% better than either.

5. Densification. 240 catalogues with HODs close to the fiducial were added in the cosmologies of the derivative grid; this alone did not stabilise the derivatives (70–170% scatter between networks).

6. Regression over pairs. Introduced as an emulator-independent estimator (Section 5.4); it revealed that the statistics respond essentially to one combination of the HOD parameters.

7. Training on catalogue differences. Halved the error of the predicted differences between held-out catalogues for ω_b and σ_8 (21%→10%, 13%→7%), down to their noise level (Table 4); the small-scale weight was dropped because it destabilised the auto-correlation forecasts. With the 192-subvolume covariance, emulator and regression then agreed on ASTRA being as good as or 0–20% better than density split (Tables 6 and 7).

Learning curve. To decide whether more training catalogues would reduce the emulator error, we retrained the emulator of the 2PCF and of the Q1, Q4 random-quartile cross-correlations on 25, 50, 75 and 100% of the training set. The median test error, in units of $\sigma_{\rm box}$, drops from 0.7–1.3 to 0.5–0.8 between 25 and 50%, but only to 0.5–0.7 at 100% (ASTRA Q1: 1.01, 0.59, 0.55, 0.51; density split Q4: 0.69, 0.51, 0.48, 0.46). The error has reached a plateau at $\simeq 0.5\sigma_{\rm box}$, comparable to the sampling noise of a single catalogue: doubling the training set ($\simeq 230$ node-hours) would improve it by $\sim 10\%$ at most, and we did not do it.

8. Covariance. With the 1786 independent boxes and the same derivatives, ASTRA's lead disappeared (Table 9).

9. Quijote. The forecasts were first computed for all six parameters. An automatic convergence test run overnight reported that the covariance was not converged, because it compared full forecasts in which the derivative noise dominates the change; once the two effects were separated, the covariance was found converged and the extra processing it had triggered was cancelled. The six-parameter forecasts were then found to depend on the derivative noise (standard and compressed estimators differing by factors 2–10), which led to the parameter subsets of Section 7.2. A first overnight result, ASTRA tighter than density split on σ_8 by a factor 2–3 and density split tighter on Ω_m, Ω_b, survives in the reliable subsets only for the auto-correlations and halo quartiles.

10. MCMC. The first inference runs on full boxes were limited by the emulator error; on one subvolume, they are limited by the prior (Section 7.4).

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